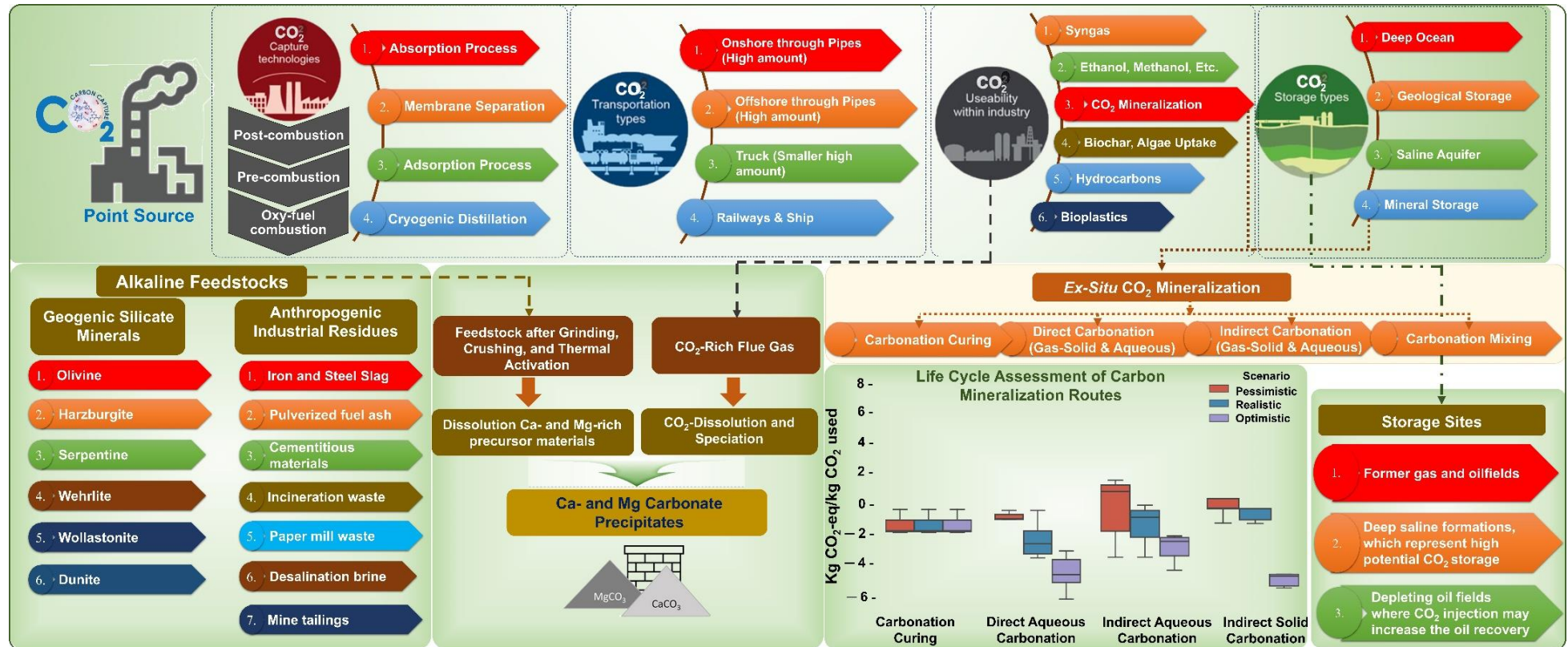


Graphical abstract



Breakthrough innovations in carbon dioxide mineralization for a sustainable future

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Abstract

Greenhouse gas emissions and climate change concerns have prompted worldwide initiatives to lower carbon dioxide (CO₂) levels and prevent them from rising in the atmosphere, thereby controlling global warming. Effective CO₂ management through carbon capture and storage is essential for safe and permanent storage, as well as synchronically meeting carbon reduction targets. Lowering CO₂ emissions through carbon utilization can develop a wide range of new businesses for energy security, material production, and sustainability. CO₂ mineralization is one of the most promising strategies for producing thermodynamically stable solid calcium or magnesium carbonates for long-term sequestration using simple chemical reactions. Current advancements in CO₂ mineralization technologies, focusing on pathways and mechanisms using different industrial solid wastes, including natural minerals as feedstocks, are briefly discussed. However, the operating costs, energy consumption, reaction rates, and material management are major barriers to the application of these technologies in CO₂ mineralization. The optimization of operating parameters, tailor-made equipment, and smooth supply of waste feedstocks require more attention to make the carbon mineralization process economically and commercially viable. Here, carbonation mechanisms, technological options to expedite mineral carbonation, environmental impacts, and prospects of CO₂ mineralization technologies are critically evaluated to suggest a pathway for mitigating climate change in the future. The integration of industrial wastes and brine with the CO₂ mineralization process can unlock its potential for the development of novel chemical pathways for the synthesis of calcium or magnesium carbonates, valuable metal recovery, and contribution to sustainability goals while reducing the impact of global warming.

Keywords: Greenhouse gases; Carbon dioxide; Mineralization; Carbon capture and storage; Sustainable carbonation processes

Abbreviations

Acid mine drainage, AMD; Carbon capture, utilization, and storage, CCUS; Carbon mineralization, CM; CO₂ capture and storage, CCS; CO₂ equivalents, CO₂-eq; CO₂ mineralization slag, CMS; Coal fly ash, CFA; Direct air capture, DAC; Enhanced oil recovery, EOR; Environmental Protection Agency, EPA; Global warming potential, GWP; Greenhouse gas, GHG; Industrial solid waste, ISW; Ion-exchange, IEX; Life cycle assessment, LCA; Phosphogypsum, PG; Rare earth elements, REEs; Stainless steel, SS; Total dissolved solids, TDS

1. Introduction

The burning of fossil fuels during industrial operations is a major source of atmospheric carbon dioxide (CO₂), which leads to an increase in the average temperature on Earth. Carbon emissions can be mitigated using renewable energy, an effective alternative to conventional fuels (Liu et al. 2024). However, replacing fossil fuels in industries that are major contributors to global CO₂ emissions, such as steel and cement production, has not yet been fully realized (He et al. 2023). For example, the cement industry accounted for approximately 7% of the global greenhouse gas (GHG) emissions (CO₂, 2.2 Gt/year) in 2014 (IEA 2018). GHGs have increased in the first two decades of the 21st century owing to the high CO₂ emissions from various sectors in China and other countries. However, per-capita CO₂ emissions are relatively high in countries with low populations, such as Qatar, Saudi Arabia, and the United Arab Emirates. Canada, the United States, and Australia show considerable per-capita emissions. In 2020, conventional fossil fuel CO₂ emissions decreased by 5.3% compared to 2019 owing to the pandemic effect; however, the emissions returned to 37.9 Gt of CO₂, which is equivalent to the 2019 pre-pandemic era (Crippa et al. 2022).

The present CO₂ concentration reached 422.11 mg/L in January 2024 (McGee 2024), with an annual rise of 2.8 mg/L (Liu et al. 2021). Reduction in CO₂ emissions is imperative, and scientists are trying to replace fossil fuels with alternative renewable energy sources such as hydrogen, solar, and wind energy. However, it is difficult to achieve the target in a short period, as global energy demand shows an upward trend, with an approximate demand higher than 29.6% by 2040 (Kramer and Haigh 2009). CO₂ capture and storage (CCS) is an imperative technology for handling climate change and securing energy security by reducing carbon emissions from point and nonpoint sources, which prevents the release of massive quantities of CO₂ into the atmosphere. Kramer and Haigh (2009) reported that high CO₂-generating industries worldwide, such as power plants and cement and steel industries, can sequester

$\sim 7 \times 10^9$ tons of CO₂ annually by 2050. CO₂ storage can be performed by mineral conversion and natural storage, such as geological and oceanic storage. The geological storage of CO₂ is widely practiced because of its large storage capability, and enhanced oil and gas recovery (Bai et al. 2016).

Carbon neutrality is a crucial measure in reducing the effect of CO₂ on the global environment and may be accomplished via carbon capture, utilization, and storage (CCUS) (Lee et al. 2023; Riahi et al. 2022). CCUS technologies have numerous potential applications in various fields, including energy recovery, food production, chemical and fuel production, refrigeration, fire suppression, and mineral formation (Qian and Han 2023; Zhou and Metivier 2023). CCS applications in heavy industries can considerably and rapidly reduce GHG emissions, as these industries are responsible for 21% of global emissions, primarily from cement, iron and steel, petrochemicals, and biofuel production industries (Page et al. 2020). To date, these approaches have demonstrated 50–68% efficiency in CO₂ capture on a small scale (Bui et al. 2018). Several studies have highlighted the importance of CCUS in reducing process-inherent CO₂ concentrations before they are released into the atmosphere (Li et al. 2023b; Ostovari et al. 2020; Tyagi et al. 2023). Various approaches, such as absorption, physical adsorption, chemical looping, cryogenic techniques, membrane gas separation, ionic liquid looping processes, biological systems, and gas hydration, can capture CO₂ (Bui et al. 2018; Chakraborty et al. 2023; Cheng et al. 2023). A simple approach is to use the entire CCS process at a single site. Additionally, there are several alternatives for capturing CO₂ from the air, followed by transportation to various storage sites using ships or pipelines (IPCC 2018; Liu et al. 2023). For example, geological shale formations are highly promising storage sites where CO₂ is entrapped in nanopores via adsorption to organic matter and clays, which have a high affinity for gas molecules (Murugesu et al. 2023).

Among current CCS technologies, carbon mineralization (CM) has drawn particular attention as a promising approach for safe and permanent CO₂ storage and waste management (Ostovari et al. 2023). Direct and indirect CO₂ reduction approaches are widely used to utilize industrial solid waste (ISW) (Liu et al. 2021), such as carbonation of ISW (blast furnace slag (390 Mt), slag from steel industries (240 Mt), waste gypsum (220 Mt), and ash from coal combustion (1000 Mt)) (USGS 2020) and the application of carbonated products in construction works to replace conventional carbon-intensive materials (Liu et al. 2024). Approximately 8% of the global CO₂ (generated due to anthropogenic activity) can be mineralized by the direct carbonation of 4.01×10^9 tons of ISW, which includes 43.5% steel slags, 16.3% cement waste, 13.5% mining waste, and 12.3% coal ashes. The carbonated materials produced by indirect carbonation to replace construction materials can reduce 3.7×10^9 tons of CO₂ emissions (Liu et al. 2021). Naturally available minerals such as wollastonite (300 Mt), serpentine (500 Mt), and forsterite (800 Mt) can be used to mineralize CO₂; however, their mining causes adverse environmental effects and is energy-intensive.

CM relies on the fundamental chemistry of ionic interactions between carbonate anions formed by the dissolution of CO₂ in water and metal cations (e.g., Mg²⁺, Ca²⁺, and Fe²⁺) to produce thermodynamically stable metal carbonate minerals as the final products (Xu et al. 2023b). This technique can potentially be widely applied in diverse environments such as brines, mafic rocks, and salt domes because of the simplicity of the reactions and ubiquity of reactant cations in nature. However, this alternative (the CM process) may pose global challenges with minimal relevance to potentially decisive mitigation options, such as institutional and regulatory barriers, capture costs and energy penalties, massive financial investments, CO₂ release risks, and environmental and health issues, if not properly understood and accepted (Park et al. 2016). Furthermore, a detailed analysis of the reaction mechanism and economics of mineralization by calculating the operating costs and simulating the mineral

chemistry of natural carbonation storage requires large-scale technical implementation (Kim et al. 2022). Therefore, technological upgrades, energy analysis, plant-level economics, and environmental analysis are necessary for the successful implementation of CCS in industrial decarbonization.

Several reviews on CO₂ mineralization have focused on CM using industrial waste (Liu et al. 2021), CO₂ recovery and utilization (Godin et al. 2021), utilization of alkaline wastes in CCS (Khudhur et al. 2022), CM using terrestrial basalts (Ostovari et al. 2023), CM mechanisms during geological storage (Kim et al. 2023), critical analysis of different CCS processes (Liu et al. 2024), and operating conditions for CM using cementitious materials (Li et al. 2024b). Here, we review the potential and significance of CM for sustainable CO₂ management and the restoration of the global carbon cycle over the next few decades. A critical analysis of contemporary technological developments in CM is presented. Additionally, efforts have been made to examine the evolution of various hands-on CCS processes to solve global CCS problems. The novelty of this review compared with the existing literature is presented in Table 1.

Table 1 Comparative evaluation of recent literature review articles published over the last three years on CO₂ mineralization with present review manuscript

References	Highlight and strength of selected review articles	Review coverage of important aspects relevant to CO ₂ mineralization (CM), its utilization, and environmental sustainability					
		Overview and mechanistic outline for CM	Various routes of CM and its field application/large-scale projects	Sustainability aspects of CM by utilizing industrial solid wastes and resource recovery	Influences and environmental threats during the CM process	Applications of CM	Outlook and challenges
(Lu et al. 2024)	- Discussed the rate of CO₂ mineralization through geochemical modeling prediction for basalt-CO₂-water interaction - Assessing the efficacy (local and global level) of CCS in basalt by reviewing the results of laboratory experiments in the literature - Discussed different types of experimental results in literature lacking accurate information regarding secondary product formation - Suggested the research requirements to fill the current knowledge gaps	- Discussed thermodynamics and kinetics of CM - Mentioned the dissolution and precipitation reactions in the basalts-CO₂-water system	- Not covered	- Not covered	- Not covered	- Not covered	- Not covered
(Liu et al. 2024)	- Discussed the importance of CO₂ capturing and mineral storage to slow down global warming - Evaluated the different methods of carbon capturing (such as liquid, solid, and microencapsulated) and <i>in-situ</i> and <i>ex-situ</i> CM	- Briefly mentioned the thermodynamics of CO₂ sorption on alkali metal oxides - Mentioned the mineral carbonation process and its field application	- Mentioned different routes of CM - Discussed different CM projects implemented at the field scale	- Not covered	- Briefly mentioned risks and challenges associated with the CM process	- Not covered	- Not covered

(Wang et al. 2024a)	- Highlighted the possible approaches for the enhancement of carbon capturing efficiency and progress of carbon storage - Discussed the current trend of technological advancement to expedite the CM process using industrial waste (fly ash) - Comprehensively evaluate the CM pathways, scale-up using novel reactor design, pretreatment of fly ash, and chemical additives	- Discussed CM reaction mechanisms and kinetics models with typical industrial waste, fly ash	- Not covered	- Not covered	- Not covered	- Briefly discussed the product formation and utilization	- Briefly mentioned the perspective of CM related to feedstock, fly ash
(Wang et al. 2024b)	- Mainly discussed CM process using iron and steel solid waste - Discussed various challenges, such as high cost, slow reaction kinetics, technical barriers, and environmental toxicity aspects associated with CM using iron and steel industrial waste	- Briefly discussed the mechanisms of CM, specifically with iron and steel slags	- Briefly discussed the different routes of CM	- Not covered	- Discussed the environmental threats, techno-economic feasibility, and scale-up	- Not covered	-Discussed the prospectives
(Zhang et al. 2023a)	- Assessed the CM potential in terrestrial basalts in different regions in China using direct injection of CO₂ gas and carbonated water - Studied the Monte Carlo simulation to assess the carbon storage capacity using terrestrial basalts	- Discussed the mineral trapping mechanisms in basalts for CO ₂ storage	- Not covered	- Not covered	- Not covered	- Not covered	- Not covered

(Sun et al. 2023)	- Discussed the mechanisms, models, applications, and characterization of CM and geological storage - Comprehensively studied the mechanisms of the complexity of long-term CM process for geological CO₂ storage - Mentioned the effects of the reservoir's environment conditions, such as pH, temperature, and pressure for CO₂ storage by mineralization - Highlighted the complexity and permeability of pore scale during mineral sealing for CO₂ storage	- Discussed the details of mechanisms and kinetics during rock, water, and CO₂ reactions for CM	- Discussed various simulation programs and experimental validations of field-scale CM for CO₂ storage - Discussed various demonstration projects for CO₂ storage	- Not covered	- Not covered	- Not covered	- Discussed the challenges and prospectives
(Ho and Iizuka 2023)	- Reviewed the mineral carbonation using seawater for carbon sequestration and utilization - Highlighted the alkaline properties of seawater due to the presence of Ca²⁺ and Mg²⁺ content as an ideal candidate CM - Assessed the research trends and recent technologies for CM using seawater as feedstocks	- Not covered	- Discussed various routes of CM primarily using seawater as feedstock (artificial and natural)			- Evaluated the potential applications of CM at a large scale	- Summarized the efficiency of seawater as a potential source of substrate for CM - Mentioned the future prospects
(Li et al. 2023a)	- Reviewed the CM efficiency using Mg-containing substrates - Discussed the latest advancement of direct and	- Not covered	- Mentioned pathways and reactions for direct and indirect routes of CM using Mg-	- Not covered	- Not covered	- Not covered	- Not covered

	indirect CM using solid and liquid-containing Mg-materials		containing minerals				
(Rashid et al. 2023)	- Reviewed the recent developments of mineral carbonation for long-term storage of CO₂ using natural feedstocks - Discussed the effects of thermal activation, soaking, seeding, and increasing reactivity by pretreatment of potential raw and waste material for CM	- Not covered	- Briefly discussed the pilot-scale plant for the CM process	- Not covered	- Not covered	- Not covered	- Not covered
(Zhuang et al. 2023)	- Reviewed the efficacy of oceanwater for CO₂ capture and conversion through mineral carbonation - Discussed the biological means (e.g., macroalgae) of carbon capture using marine chemical carbon sink at high alkalinity by adding inorganic carbon for CO₂ sequestration - Explored the effect of increasing weathering and alkalinity and materials in the regulation of carbon sink for carbon storage	- Not covered	- Not covered	- Not covered	- Not covered	- Not covered	- Not covered
(Stokreef et al. 2022)	- Discussed the candidate material, carbonation modeling, case studies, and reaction mechanisms and kinetics of mineral carbonation using ultramafic tailings	- Discussed in detail the carbonation reaction, mechanisms, and thermodynamics using ultramafic tailings as a substrate	- Discussed in detail various routes of the CM process	- Not covered	- Not covered	- Not covered	- Briefly mentioned the prospectives
(Hanifa et al. 2023)	- Reviewed the contemporary	- Not covered	- Discussed different routes of	- Not covered	- Not covered	- Not covered	- Briefly discussed the

	developments and commercialized technologies for CO ₂ capture and storage, primarily using solid waste from cement industries		CM from cement-based materials, by-products from cement industries - Critically discussed the detailed mechanisms and flow schemes of various commercial approaches for CO ₂ storage linked with cement industries				challenges and perspectives for CM using cementitious materials
In present study	- Discussed the overview and mechanistic outlines of CO₂ mineralization and its different routes - Mentioned the current status of CO₂ mineralization following patents and publications - Analyzed different CO₂-capturing technologies and integrated mineral carbonation for utilization of different wastes for resource recover - Critically analyzed the environmental threats during mineral carbonation	- Mentioned CO₂ mineralization overview and chemistry	- Critically discussed different carbonation routes - mentioned the current status of pilot and field-scale projects for the CM process	-Sustainable management of different wastes for their utilization and as feedstocks and valorization	- Covered the environmental concern of the CM process	- Discussed different applications of the CM process	- Discussed outlook and challenges associated with the CM process

2. Methodology for systematic literature review

Several review manuscripts related to CO₂ storage through the CM process are available in the literature, which discuss many objectives, such as pilot-scale studies (Hanifa et al. 2023), potential feedstock application for CM (Stokreef et al. 2022), elaborating the potential of seawater for CM (Ho and Iizuka 2023), and utilization of specific feedstocks or ISW (Ca-or Mg-containing natural raw materials, ultramafic tailings, fly ash, iron, and steel slags) (Li et al. 2023a; Stokreef et al. 2022; Wang et al. 2024a; Wang et al. 2024b). This review comprehensively describes recent developments in innovative technologies in CM and their current status in large-scale industrial implementation. It deals with a general mechanistic outline of the CM process and critically discusses the advantages and limitations of different CM process routes for long-term CO₂ storage. Furthermore, the utilization of various industrial byproducts or solid wastes, resource recovery through mineral carbonation, applications of the CM process, and the sustainability and environmental threats of CM approaches are discussed. This was performed by a thorough literature search using different online databases, such as Scopus, Science Direct, Google Scholars, SciFinder, ResearchGate, Wiley, SpringerLink, Web of Science, and Espacenet (patent database), to obtain appropriate journal and patent publications in the last seven years. To obtain a wide range of searches, several keywords such as GHGs, CO₂, Mineralization; Carbon capture and storage, sustainable carbonation processes, CO₂ mineralization, mineral carbonation, mineral trapping, brine valorization, green concrete, CO₂ mineralization slag (CMS), industrial by-products, and life cycle assessment (LCA) were used to obtain relevant research documents. We selected more than 250 articles and patents and carefully reviewed them. Only those deemed relevant, noteworthy, and impactful were selected and mentioned in this review. Most of the selected articles were published within the past five years to ensure an updated assessment of the most recent and advanced technological

developments. A detailed analysis of the selected articles helped prepare a table of contents for this manuscript to present the considerably different content available in the literature.

3. CO₂ mineralization

3.1 Overview

The CM technology of the CCS process is versatile and can mitigate CO₂ emissions from industrial sources on a large scale. CM is a versatile and thermodynamically downhill route (i.e., a Gibbs free energy change of <0) that can ensure long-term and environmentally benign atmospheric CO₂ fixation, while reducing the natural CO₂ conversion timescale from several years to a few hours ([Gadikota 2021](#); [Liu et al. 2023](#)). Seifritz ([1990](#)) proposed CM in the early 90s to sequester anthropogenic CO₂ and was subsequently evaluated by Lackner and co-workers ([Lackner et al. 1995](#)). CM technologies are based on the reaction of CO₂ (gas, liquid, or dissolved in water) with Ca and Mg ions in various host media (e.g., rocks and sediments) to transform them into stable carbonate minerals (e.g., magnesite and calcite). Although this review focuses on the above-discussed physicochemical reactions for CO₂ conversion owing to their commercial viability, it is worth mentioning that biological routes play major roles in nature. For example, 1) CO₂-fixing bacteria can adsorb and transform CO₂ into carbonates ([Qian et al. 2022](#)), 2) microalgae can facilitate the conversion of CO₂ into valuable biochemicals ([Daneshvar et al. 2022](#)), and 3) carbonic anhydrase can boost CM ([de Oliveira Maciel et al. 2022](#)).

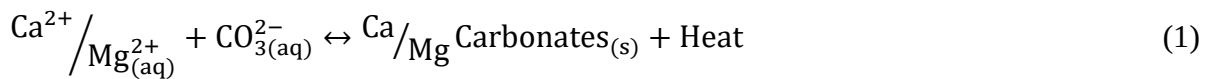
The major advantages of the CM process are as follows: host mineral resources are available globally; this process offers a permanent solution for CO₂ fixation by binding CO₂-containing fluids and gases to solid igneous rocks (typically basalt or peridotite minerals), which is an exothermic reaction that requires no energy input, and is a cost-effective route for CO₂ fixation. Apart from these advantages, there are a few shortcomings of using CM. According to Park and Fan, gas-solid interactions cannot always be used to effectively create

stable metal carbonates (c.f., kinetics-controlled formation of metastable hydrated carbonate phases such as nesquehonite) because the distribution of optimal mineral resources is only superficially understood (Park and Fan 2004). Therefore, it is crucial to understand the fundamental basis of CM reactions and the practical routes through which this technology has been applied to various systems.

CM is thermodynamically stable. Theoretically, it can permanently fix atmospheric CO₂ under ambient conditions because the process has a lower energy state than that of the reactants (CO₂ and silicates). However, it is possible that mineralized carbon can dissolve in the presence of strong acids under ambient conditions. Therefore, there may be a risk of CO₂ release into the atmosphere if precipitated carbonates are exposed to strong acids (Allen and Brent 2010). Teir et al. (2006) reported the effect of pH on the CM process and revealed that carbonated minerals are unstable in nitric acid environments at various concentrations. The dissolution of the Mg and Ca fractions from their respective carbonates in separate solutions (initial pH of 1) after several days of stabilization was 9%, whereas, at an initial pH >2, the fraction of dissolved minerals in the solution was <1%. Infrared analysis of the reactor atmosphere revealed a more rapid release of CO₂ from CaCO₃ than MgCO₃. The release of CO₂ gas was approximately 1.5% from CaCO₃ and 0.0% from MgCO₃ at pH 1. The release of CO₂ from both carbonates was not detected at pH 2. Similarly, temperature and humidity modulate the stability of carbonated CM products in the natural environment. Dissolved Ca ions readily precipitate from the reactive aqueous fluid at <280 °C once calcite and/or aragonite is supersaturated, whereas dissolved Mg ions precipitate as carbonate magnesite and dolomite at ~80 °C or higher (Saldi et al. 2009). At lower temperatures, the precipitation of these minerals is kinetically inhibited.

3.2 General mechanistic outline

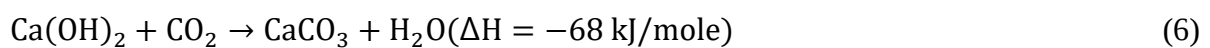
Equations (1)–(4) describe the CM process. CM occurs via the reaction of water containing dissolved CO₂ with metal cations in various media to form solid metal carbonates (Equation (1)), and the individual reaction steps are considerably influenced by the solution pH (Demirbas 2007; Lee et al. 2023). Equations (2) and (4) indicate that the formation of carbonate ions (CO₃²⁻) is favored at an alkaline pH. The most accessible substance for moving the pH toward alkalinity is sodium hydroxide (caustic soda).



This process relies on the sources of Ca²⁺ and Mg²⁺ ions that interact with CO₃²⁻. Saline environments have immense potential for applications in CM technologies. In particular, natural and synthetic brines have been identified as appropriate media for CM because of their high Ca and Mg content (Liu et al. 2019; Power et al. 2017). Under typical brine conditions, CaCO₃ forms more readily than MgCO₃ because of its greater thermodynamic stability (Ji et al. 2022) and faster kinetics for the dehydration of Ca than Mg during nucleation and crystal growth. In addition, metal cations can be extracted from geological materials. For example, mafic and ultramafic rocks contain large amounts of Mg and Ca silicate minerals (e.g., forsterite, diopside, and Ca-rich plagioclase). However, metal extraction from these materials commonly requires high acidity, which poses considerable environmental risks.

Simple oxide and hydroxide phases, such as CaO and Ca(OH)₂, are considered sources of metals for CM, as shown in Equations (5-6). The efficiency of these processes is mainly determined by the surrounding relative humidity, temperature, and surface charge of the particles (Montes-Hernandez et al. 2012). The CM rate typically increases with increasing

relative humidity because thicker water films formed on solid surfaces can have higher contents of dissolved CO₂ and Ca²⁺ (Murugesu et al. 2023). Temperature considerably affects the carbonation rate, as expected from the Arrhenius equation (Stokreef et al. 2022). Physically, an increase in temperature facilitates the dehydration of both the substrate surfaces (e.g., the surfaces of Ca(OH)₂ particles) and hydrated ions, which increases the reaction rates (Lackner et al. 1995).



3.3 CO₂ mineralization routes

CO₂ and feedstocks containing metal ions are required for *ex-situ* mineral carbonation to store CO₂ as a thermodynamically stable carbonate mineral. Point sources of CO₂ include flue gases from steel, cement, and electricity manufacturing units (Lux et al. 2018). Some carbonation processes use flue gases directly, whereas others require pure CO₂. During the initial stages of CM, Ca, Mg, and Na generally exist as dissolved ions or primary (hydr)oxides (Franks et al. 2023). In the subsequent phase, these intermediates react with dissolved CO₂ to form CaCO₃, MgCO₃, or NaHCO₃. Direct and indirect carbonation can be considered production processes for carbonate compounds (Wang et al. 2024b), which have additional uses in various contexts (Oh et al. 2019). Conversely, mixing and carbonation curing are discretionary steps in concrete production as a carbon sequestration medium.

3.3.1 Direct carbonation

Direct carbonation refers to a process in which metal dissolution and carbonate precipitation co-occur in the same reactor, or there is only one pathway for the reaction (Li et al. 2023c). Dry gas-solid (moisture <0.2) and aqueous mineral (gas-liquid and gas-liquid-solid, i.e., slurry) carbonation are two general routes for achieving direct carbonation. Direct gas-solid carbonation is an approach in which gaseous CO₂ is injected into a reactor to immediately

interact with the ground feedstock (host mineral cation source) (Ho et al. 2020). Direct carbonation pathway is the simplest carbonation process, and smooth heat recycling from an exothermic reaction is a major advantage (Zevenhoven et al. 2008). In addition, it allows for less consumption of acids or bases and a low water footprint during direct solid-gas carbonation for the safe sealing of CO₂ in the solid phase while directly capturing and separating CO₂ from the gas phase by utilizing minerals and solid wastes (Moon et al. 2024). Minerals, such as wollastonite, serpentine, and forsterite, react spontaneously with CO₂ in the dry route of CM with Gibb's free energies of -44.6, -16.9, and -43.0 kJ/mol, respectively (Benhelal et al. 2020). However, this process is kinetically slow and has a low conversion rate, which results in poor commercial viability. Several approaches have been proposed to increase the direct carbonation rate. For example, *in-situ*, dry and moist (10% water) gas-solid carbonation was performed using chrysotile (a type of asbestos for Mg-source) in the presence of CO₂ (67 mol%) at ambient pressure, thermal preconditioning (at 300 to 1200 °C), and doping with alkali metal ions. The presence of water, cesium as a trace alkali metal, and a temperature of 375 °C boosted mineral carbonation to form MgCO₃ by up to 2.7 times (Larachi et al. 2010).

Direct aqueous carbonation is the most well-understood technology for gas-liquid-solid multi-phase reaction systems and provides favorable conditions for carbonation (Sanna et al. 2014). Direct aqueous carbonation requires CO₂ conversion into a carbonic acid, H⁺, and CO₃²⁻ solution after dissolution in an aqueous system, followed by a reaction with Ca²⁺/Mg²⁺ ions (released from dissolved primary or secondary minerals, e.g., silicates or (hydr)oxides) to form carbonate precipitates (Miao et al. 2023; Tao et al. 2021). Hence, it allows three co-occurrences in the reaction vessel: the development of a mildly acidic environment owing to the dissolution of CO₂ in water to form HCO₃⁻, the leaching of Ca/Mg from solid matrices (minerals or industrial wastes), and the precipitation of Ca/MgCO₃ (Olajire 2013). This process exhibits a better reaction rate and carbonation efficiency than direct gas-solid carbonation because of the

higher degree of mass transfer. Water is used as a solvent to dissolve CO₂ and facilitate the ionization of Ca or Mg for rapid solid carbonate formation (Bacocchi et al. 2016). However, the dissolution of the Ca/Mg-bearing feedstocks acts as a rate-limiting step during the precipitation of ions into carbonates. Therefore, the use of several types of additives (e.g., NaCl, NaHCO₃, and Na₂CO₃) and key reaction parameters (such as pH, solid/liquid ratio, operating temperature and pressure, particle size, and CO₂ concentration) must be optimized to improve the kinetics of silicate dissolution (Bacocchi et al. 2010; Poletini et al. 2016). The addition of carbonate-bearing additives, such as Na₂CO₃ and NaHCO₃, can maintain the pH of the solution and increase Ca²⁺/Mg²⁺ leaching from the substrate by producing more H⁺ ions in the solution to enhance the overall carbonation efficiency (Ji et al. 2017). However, it has a slightly lower net avoided greenhouse warming potential (384 kg of CO₂ per MWh_{el}) than dry gas-solid carbonation (473 kg of CO₂ per MWh_{el}), owing to the higher consumption of material and energy (Ghasemi et al. 2017).

3.3.2 Indirect carbonation

An indirect carbonation process is associated with two or more consecutive reaction steps, that is, the extraction of the Ca/Mg reactive components using chemicals at a low pH during the initial step and their reaction with CO₂ to form carbonates during the final step under alkaline conditions in different reactor vessels (Wang et al. 2022). Hence, a change in pH is required during indirect carbonation. Ca²⁺ and Mg²⁺ ions from mineral-rich host rocks were leached out under acidic conditions during the first step. Simultaneously, undesired products such as oxides/hydroxides can precipitate under basic conditions during the subsequent steps. The effective application of this route depends on controlling the solution pH, which directly controls the reaction rates of individual processes (Azdarpour et al. 2015). Silicate dissolution is promoted at low pH values, whereas carbonate precipitation is favored at high pH values (Ambarita et al. 2024). Therefore, indirect carbonation is performed in three separate steps:

acid ion extraction, gas-solid carbonation, and pH swing (Pacala et al. 2018). During acid ion extraction, acidic solutions (introduced before the carbonation reaction) can improve the reaction rate along with the leaching of $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions from their host rocks by polarizing and weakening the bonds within their structure. For example, inorganic acids, such as hydrochloric or sulfuric acid, have been tested for their ability to extract Mg^{2+} ions from serpentine (Lackner et al. 1995; Maroto-Valer et al. 2005). However, their high energy consumption and potential environmental impact limit their applicability.

Organic solvents and chelating additives were tested for their ability to promote metal extraction. The economic and ecological feasibility of these processes can be questioned, as the recycling and disposal of additives is a challenge. During indirect gas-solid carbonation, the mineralization of the Ca/Mg-O and Mg/Ca(OH)_2 phases is generally extremely rapid and efficient (Moon et al. 2024). An additional step was required for the production of the relatively reactive Mg/Ca(OH)_2 by the hydration of Ca/MgO . However, utilization of Mg/Ca(OH)_2 can expedite the reaction kinetics compared to Ca/MgO during solid carbonate formation at similar pressure (>10 bar) and temperature (>500 °C) (Zevenhoven et al. 2008). Indirect carbonation yields a high-purity byproduct compared to the direct carbonation process under modest reaction conditions (potentially with the application of a modified pH solution) (Liu et al. 2023; Romanov et al. 2015).

4. Sustainable technologies for CO_2 capture, storage, and applications

The development of CCS technologies at point sources of CO_2 emissions is a recent strategy implemented worldwide to capture CO_2 , followed by its sequestration, direct utilization, and conversion into useful chemicals or fuels (Li et al. 2016).

4.1 Current status of CSS processes: Patents and publications

CO_2 is an inert gas with no combustion heating value; however, it is a crucial environmental concern because it is one of the major GHGs (Moon et al. 2024). Different technologies have

been documented in journal articles and patents for capturing CO₂ for storage or valuable applications (Yu et al. 2023). CCS has emerged as a leading-edge technology for alleviating GHG emissions from large industrial facilities, oil refineries, and fossil fuel-based power plants (Khosroabadi et al. 2021). This can expedite a safer stabilization process for atmospheric GHG concentration levels and help maintain fossil-based fuels within the energy matrix. The number of patent filings and papers covering capture technologies has increased annually, following the typical exponential trend observed for emerging technologies (Khosroabadi et al. 2021). Until 2012, more than 1,000 patents were granted for solvents, sorbents, and membrane-based applications, of which 60% were granted after 2000 (Li et al. 2013).

According to Espacenet (patent database), there are more patents and scholarly publications on absorption and adsorption than other capture techniques. Although there are only a limited number of researchers working on enzyme-based capture of CO₂, as reflected by fewer articles/patents published each year, this is a technological approach with tremendous potential (Quintella et al. 2011). The lack of patents/papers might be due to a lack of motivation in researchers with relevant expertise in enzymes for CO₂ capture and the challenges associated with manipulating enzymes in the laboratory. Moreover, adsorption yields better CO₂ capture results because it involves sophisticated technology. The primary contributors to adsorption include polymers, zeolites, activated carbon, molecular sieves, silica, and metal-organic frameworks (Khan et al. 2023). However, enzymes are still in the initial stages of research and require new technological advancements to be practical for use in commercial plants. Hybrid technologies such as membrane-integrated thermodynamics, adsorption and absorption processes, and absorption and adsorption methods documented in patents are evolving rapidly, spanning the integration of conventional areas of CO₂ collection (Quintella et al. 2011).

Technologies for CO₂ mineralization for carbonate formation methods for CO₂ reduction are relatively mature compared with chemical methods. Fig. 1 a shows the overall

annual journals and patent publications for the keywords searched for CO₂ mineralization processes in Scopus - Document search and Espacenet-Worldwide patent search, respectively. Documents (journals and patents) referred to CO₂ mineralization were published rapidly starting in the early 2001s, slackened in the 2010s, and again showed an increasing publication rate. The steady growth in research output can be attributed to global efforts that have triggered the necessity of reducing atmospheric CO₂. However, a lack of support for CO₂ mineralization projects, small investments, and proper economic incentives eventually stabilize publications and patents. This leads to more journal publications compared to patents (Fig. 1 a). The grouping of publications for journals (Fig. 1 b) according to the country/region or affiliation of the first author or patent assignee indicates that 12 countries have published a maximum number of journals, with China outnumbered in journals.

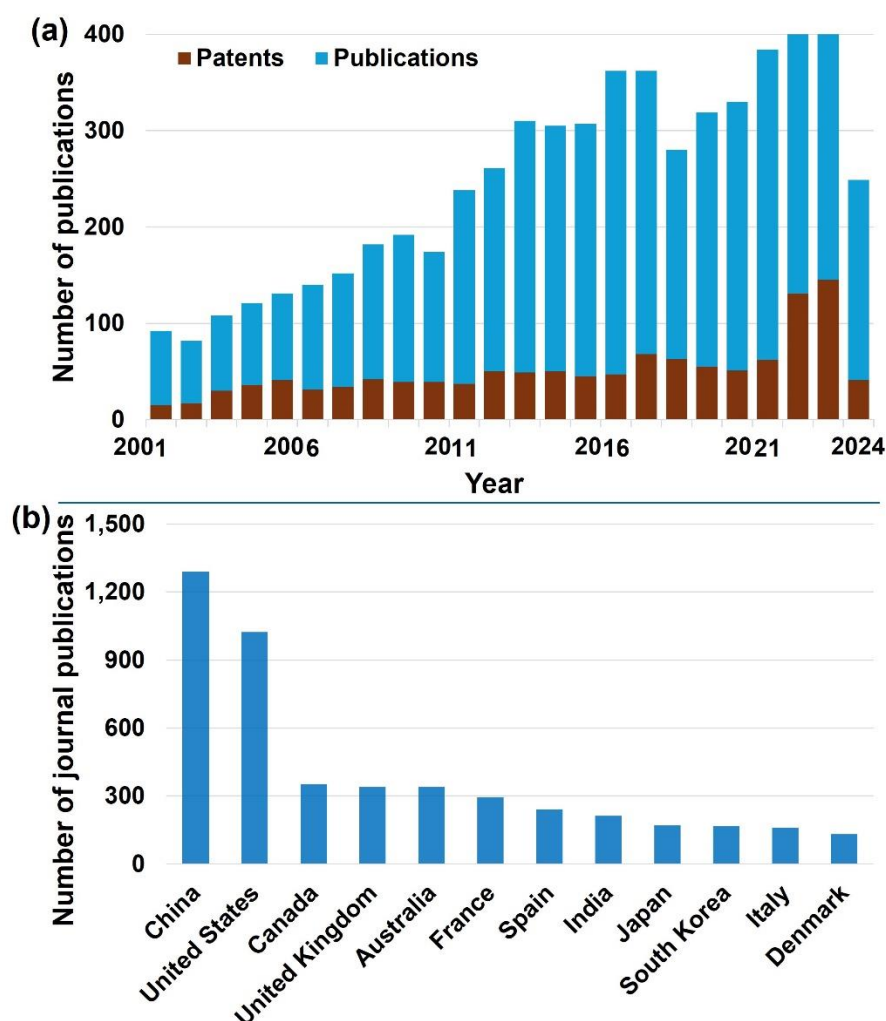


Fig. 1. Trends of publications and patents worldwide related to CO₂ mineralization, (a) global publications (journals and patents) counts, (b) total number of journal publications in top 12 countries (Year- and country-wise journal and patent publications were obtained from Scopus - Document search and Espacenet-Worldwide patent search, respectively (data from 2001 to June 6, 2024))

4.2 CO₂ capture technologies practiced

CO₂-containing polluted gas streams are emitted during energy conversion pathways such as anaerobic digestion, combustion, gasification of carbonaceous resources (e.g., coal, plastics, biomass, and natural gas), and manufacturing industries (e.g., cement, steel, fertilizer, fuel cell, and paper and pulp) (Gadikota 2021). The CO₂ separation process may contribute 70–80% of the total CCUS process, depending on the technology selection that leads to CO₂ formation during combustion and its types (Raza et al. 2019). Conventional processes, such as adsorption, cryogenic separation, solvent-based absorption, and membrane-based separation, are practiced to separate CO₂ from its mixed gas streams generated from post-combustion. The feasible and reliable CO₂ transportation from its capture to storage locations is also crucial for an efficient CCS process, which includes high-pressure pipelines (onshore and offshore), ships, trucks, and railways (Svensson et al. 2004). Fig. 2 shows schematic diagrams of fuel combustion-based industries integrating carbon capture and mineralization using different alkaline waste feedstocks (Li et al. 2016).

In the 1970–80s, CCS facilities were used in industries such as fertilizer and natural gas processing, where CO₂ separation was a routine process to fulfill the demand for CO₂, mostly in enhanced oil recovery (EOR) and to a lesser extent in the decaffeination process, fertilizer, food preservation, beverage carbonation, fire suppressants, and pharmaceuticals (Liu et al. 2018a). However, innovative reactor designs, economical process schemes, and advanced materials are required to achieve successful CO₂ capture. CO₂ capture from the natural gas and fertilizer industries has been practiced for decades and has recently been operational in coal-based power industries. To prevent CO₂ from being released into the atmosphere, its capture/separation from industrial exhaust gases is the primary step in CCS. Amine absorption and calcium looping processes have been installed with natural gas-based power units to capture CO₂, which could reduce carbon emissivity by 82–87% compared with typical power

plants (Strojny et al. 2023). The major carbon capture processes are pre-combustion, post-combustion, and oxy-fuel processes (Strojny et al. 2023; Wang et al. 2020).

Pre-combustion requires fuel conversion steps, such as the use of a gasifier under a low-oxygen atmosphere for the conversion of fuels into syngas, that is, a mixture of H_2 and CO , which subsequently undergoes a water-gas shift reaction in a catalytic reactor, where syngas further reacts with steam (steam reforming) or oxygen (partial oxidation) to form more H_2 and convert CO into CO_2 (Godin et al. 2021; Leung et al. 2014). The conversion of CO into CO_2 by steam reforming or partial oxidation followed by water-gas shift reactions results in 15–60% of the CO_2 in the syngas, which can be separated by physical adsorption or chemical absorption to achieve an H_2 -rich stream (Rackley 2017). Furthermore, smaller and more compact equipment containing different types of solvents can be used to capture concentrated and pressurized CO_2 , serving as a relatively low energy-intensive process compared to a post-combustion system (Godin et al. 2021). The high capital investment required to develop a fuel conversion facility is a major limitation to pre-combustion carbon capture. A coal-based power plant integrated with gasification-combined cycles can capture pre-combustion CO_2 , although its application in power plants is challenging owing to a 7–8% efficiency loss. However, energy loss was improved by the US (Department of Energy and Electric Power Research Institute) by enhancing the efficiency of gasifiers (GCCSI 2012).

In post-combustion capture, CO_2 is separated from flue gas (containing 4% to 15% of CO_2 along with NO_x , SO_x , water, oxygen, and inert gases) after the complete combustion of fossil fuels (Al-Mamoori et al. 2017). The post-combustion process is the preferred option for implementation in existing power plants, with a CO_2 recovery rate of up to 800 tons/day (Wall 2007). However, low CO_2 concentrations in flue gas require separation prior to any application, which requires large equipment and high capital investment to handle massive volumes of flue gas (Olajire 2010). Additional separation processes, such as absorption, adsorption, cryogenic,

and membrane-based methods, are required to recover high-purity CO₂ (>95.5%) from combustion exhaust gases because of their high parasitic loads and low CO₂ content (4% for gas-fired and 7–14% for coal-fired). The post-combustion capture equipment was placed immediately after the conventional industrial purification system to decrease the number of contaminants and avoid the degradation of solvents used for CO₂ capture. The design and implementation of the equipment pose major challenges owing to the unfavorable conditions of flue gas. For example, low CO₂ partial pressure results in a low driving force for its capture. This results in the requirement of a robust separation process, such as absorption by an amine solvent with a high energy requirement for desorption and solvent regeneration ([Al-Mamoori et al. 2017](#); [Chao et al. 2021](#)). Flexible operation in power plants to capture CO₂ at peak hours during high electricity charges and string solvents for regeneration at off-peak hours can help reduce overall energy costs ([Moioli and Pellegrini 2019](#)). According to the International Energy Agency report, in 2022, approximately 45 Mt of CO₂ will be captured per year from 35 commercial CCS units installed in various industries, which will increase to 200 facilities for CCUS with the capacity to capture 220 Mt of CO₂ per year by 2030 ([Subramanian and Madejski 2023](#)). According to the US National Energy Technology Laboratory, approximately 32% and 65% of electricity consumption costs increase for CO₂ capture from post-combustion exhaust gases in gas- and coal-based thermal power plants, respectively ([Kanniche et al. 2010](#)). A post-combustion CO₂ capture analysis was performed using a CCS process integrated with a gas power plant to achieve negative CO₂ emissions ([Subramanian and Madejski 2023](#)). CO₂ was absorbed by an amine solvent (30% monoethanolamine) from the flue gas generated during fuel combustion in a gas turbine and passed through a heat-recovery steam generator for electricity generation.

In oxy-fuel combustion, the fuel is combusted using nearly pure oxygen instead of air to obtain water vapor, particulates, SO₂, and CO₂ from exhaust gas ([Kim et al. 2022](#)). The

generated flue gas is passed through a combustion chamber to control flame temperature. A large quantity of CO₂ is separated, compressed, and transported after the condensation of water vapor, making this a cost-effective and less energy-intensive process ([Seddighi et al. 2018](#)). Electrostatic precipitation and desulfurization methods are used to remove particulates and SO₂, respectively, to obtain 80–98% pure CO₂ based on the type of combustion fuel used ([Yadav and Mondal 2022](#); [Zero 2013](#)). Furthermore, this process reduces the NO_x content and volume of the exhaust gas, thereby decreasing the desulfurization cost prior to water vapor condensation. The high operating cost owing to the consumption of pure oxygen (obtained from an energy-intensive air separation process) makes the process 7% more expensive. This process has an energy penalty compared to a plant without a CCS facility ([Burdyny and Struchtrup 2010](#); [GCCSI 2012](#)). Additionally, a high concentration of SO₂ in flue gas can increase the corrosiveness of the system, which limits the implementation of high-capacity (1000–2000 MW) oxyfuel-fired projects, except for a few coal-based projects with capacities of 25–250 MWe ([GCCSI 2012](#)).

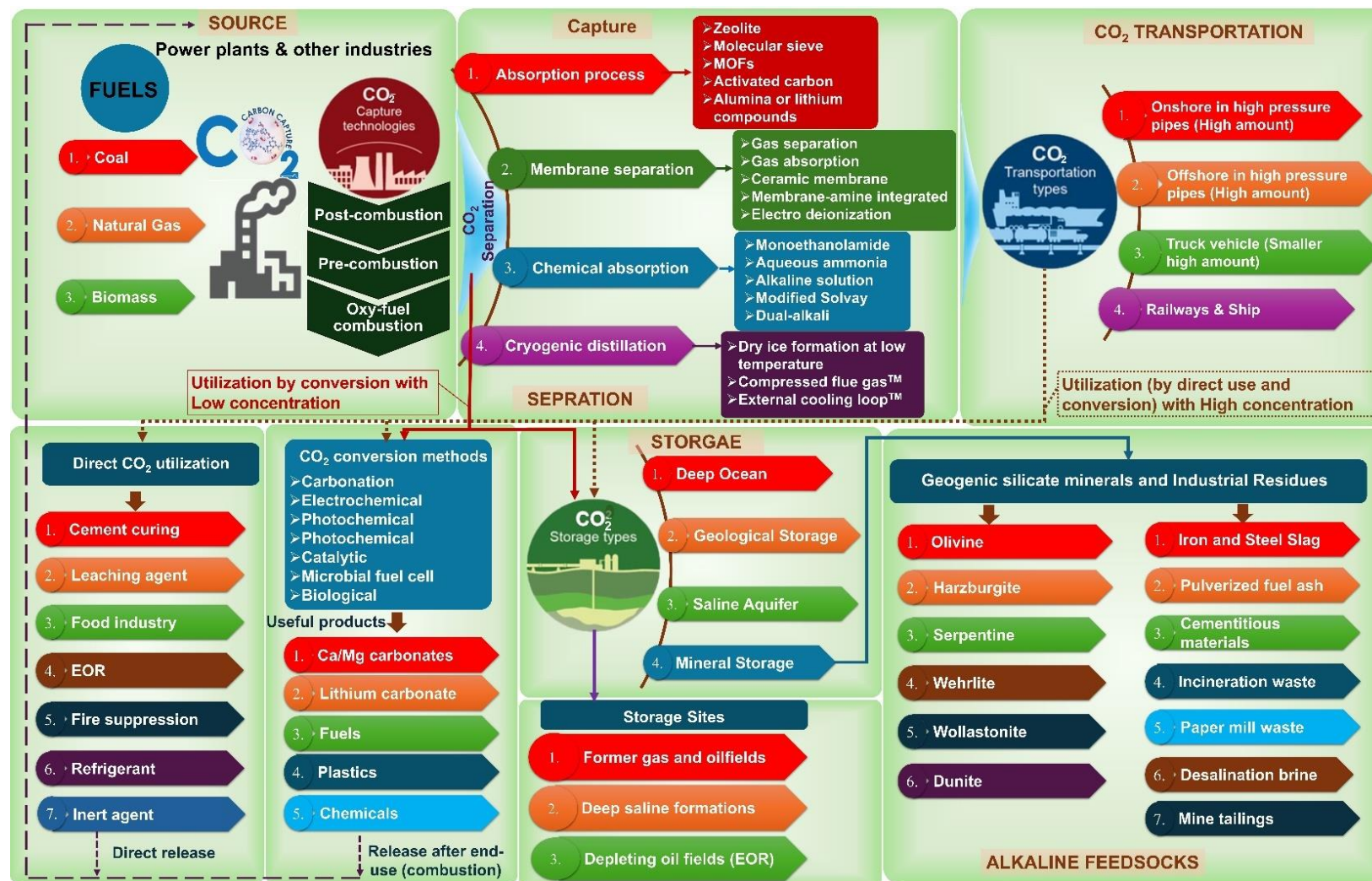


Fig. 2. Schematic diagram of CO₂ emission from fuel utilization in power plants or other industries followed by its capture, transportation, and conversion/utilization (Adapted from (Li et al. 2016))

4.3 Integrated mineral carbonation for resource recovery from waste effluents

Integrated mineral carbonation is a process in which industrial wastes, such as waste rocks, brine solution, fly ash, and mine tailings, are used as feedstocks for CO₂ mineralization.

4.3.1 Recovery of Ca and Mg from the brine solution

Brine wastewater is a saline waste solution with total dissolved solids (TDS) formed during industrial processes, such as oil and natural gas production (oil-field brines) and desalination plants (Harutyunyan 2014). Modern desalination plants rely heavily on nonrenewable energy sources, and the released CO₂ contributes considerably to global warming. The processes of receiving saltwater, treating and demineralizing it, disposing brine, and discharging treated water are energy-intensive steps in a desalination plant. The energy production chain can be simplified because of the proximity of seawater desalination facilities to fossil fuel power plants (Bang et al. 2019; La Plante et al. 2021). These brine solutions can serve as liquid components during carbonation. High concentrations of Ca and Mg in desalination brine are promising components of CM. However, Mg²⁺ and Ca²⁺ interfere and compete with each other during carbonate precipitation, resulting in a low conversion ratio of CO₂ to carbonate. Bang et al. (2019) reported an improvement in the CO₂ conversion efficiency from 12% to 69% through the use of sequential CM of Ca and Mg using 15% CO₂ to avoid the competitive and hindrance effects of Mg on the formation of CaCO₃ nuclei. A novel method for CM using spent CaCl₂ solutions involving combined reaction-extraction-crystallization processes was proposed by Dong et al. (2018). Through titration, the effects of Na⁺, K⁺, Mg²⁺, Al³⁺, NO₃⁻, and SO₄²⁻ ions present in distiller waste could affect the CaCl₂ conversion rate and the crystalline structure of the final CaCO₃ product. The authors verified this experiment using simulated distiller waste and evaluated it as an essential step toward scaling up (Dong et al. 2018). A high-TDS brine (>120 g/L) obtained from an oil and gas extraction plant was valorized by treating it with fly ash (enriched in CaO) to increase the pH to 9 and form CaCO₃

by sequestering CO₂ at a pressure of 14 bar for a reaction time of 2 h (Soong et al. 2006). This process can recover Ca (as CaCO₃) from brine and fly ash. The produced minerals, predominantly CaCO₃ or MgCO₃, can be used in various industrial applications, such as construction materials, abrasives, and fertilizers.

4.3.2 Recovery of lithium using mineral carbonation

Chen et al. (2017) presented a novel coupling technique for CO₂ mineralization and solvent extraction to precipitate Mg from brine with a high Mg/Li ratio. To extract Mg from brine, an organic amine was used to remove the HCl produced during the CO₂ mineralization process and accomplish the continuous conversion of MgCl₂ to MgCO₃. Optimal conditions allowed for a maximum conversion of 67.41% of Mg and a reduction in the Mg/Li ratio of the raffinate from 20 to 5.4 for smooth recovery of Li. To maximize Ni sulfidization and CM simultaneously, Wang et al. (2021) explored the possibility of using *ex-situ* direct aqueous CO₂ mineralization of pure olivine. By supplying a gas combination of 95% CO₂ and 5% H₂S, CO₂ mineralization in olivine resulted in the release of Ni²⁺ and Co²⁺ from the silicate olivine. In addition, tests were performed on genuine tailings from a Minnesota copper-nickel-sulfide mine to further examine CO₂ mineralization and boost metal recovery (Wang et al. 2021). The findings showed that CO₂ mineralization can be applied to ultramafic mine tailings with improved metal recovery.

Continuous efforts are being made to increase the product value of CM and investigate its economic viability. During lithium precipitation, CO₂ is a viable source for carbonation, replacing other carbonate sources, such as K₂CO₃ and Na₂CO₃, to produce Li₂CO₃ (Kim et al. 2024). CO₂ gas is dissolved in a Li-containing aqueous solution to form H₂CO₃ at a pH of 6.3, which reacts with Li to produce soluble and ionizable LiHCO₃. Subsequently, at pH >8, carbonate species dominate and precipitate to form Li₂CO₃ (Kumar et al. 2023). In addition, CO₂ acts as a leaching agent for the selective recovery of lithium (by forming carbonic acid)

followed by the precipitation of Li_2CO_3 from the spent LiFePO_4 cathode materials. Using CO_2 as a leaching agent can sequester 120 kg CO_2 /ton of spent LiFePO_4 batteries while consuming 2.29 MJ of energy and reducing GHG emissions by 194 g of GHGs with a revenue of 4.04 USD per kg of LiFePO_4 cells (Xu et al. 2023a). CO_2 mineralization and simultaneous extraction of valuable metals from different wastes through integrated mineral carbonation are currently under development. However, economic feasibility should be improved, and process advancement must be upgraded using new and sustainable technologies that present opportunities for innovation across various industrial sectors to reduce CO_2 emissions.

4.3.3 Recovery of rare earth elements from acid mine drainage

Using alkaline wastes such as acid mine drainage (AMD) and mine tailings from mining industries for CM to yield sustainable Ca/MgCO_3 can help stabilize CO_2 and recover rare earth elements (REEs) (Lee et al. 2016). Remediation of mining effluent is commonly performed by adding lime to neutralize the solution, making AMD a rich source of Ca, including other REEs (Zhang et al. 2023b). Lee et al. (2016) applied the CO_2 injection method during the neutralization of an AMD solution to precipitate CaCO_3 with an estimated carbonation efficiency of 0.54 g of CO_2 per kg of AMD. CO_2 sequestration using the AMD neutralization method without any pretreatment or additional unit operation at ambient temperature and pressure over a short operation time was evaluated as a sustainable method. Energy requirements and techno-economic feasibility studies have been conducted for mineral carbonation using flue gas CO_2 from an industrial plant and Mg-containing minerals obtained from serpentine-based mine tailings (Pasquier et al. 2016). An energy of 7.8 GJ per ton of CO_2 is consumed to store 234 kg of CO_2 using one ton of mine-tailing serpentine rock, as estimated by modeling the laboratory-based experimental results. The operating process cost of 144 USD per ton of CO_2 , with a total revenue of 644 USD per ton of CO_2 (income generated by by-

product sales and carbon credit tax), was estimated in a 1.4-year payback period ([Pasquier et al. 2016](#)).

The electrolytic carbonation approach can use different waste effluents such as brine electrolytes, fly ash, and CO₂ to produce high-quality CaCO₃. [Lu et al. \(2016\)](#) reported a 32.4% more dissolution of fly ash in a brine solution to liberate Ca during electrolysis, which was mineralized into CaCO₃ using CO₂ with a capture efficiency of 18.42 kg-CO₂ per ton of fly ash and an energy consumption of 29.3 kJ per mol of CO₂. [Vaziri Hassas \(2020\)](#) investigated a novel, eco-friendly, staged precipitation process using CO₂ mineralization to recover REEs and critical elements during AMD treatment. Precipitation studies were performed using NaOH and CO₂/NaOH to determine the viability of staged precipitation for recovering REEs from AMD. When NaOH was used, REEs precipitated in a pattern similar to that of the tetrad classification of lanthanides. Approximately 70% of the REEs precipitated at the desired AMD treatment pH, with the remaining 30% released with the treated water. At pH values below seven, the CO₂ mineralization mechanism recovered 90% of the Al and over 85% of the REEs, while preventing the precipitation of most of the Fe (i.e., 65%). In contrast, the REE carbonate precipitation pattern during CM corresponded to the trend in the hydration energies of these elements.

4.4 Green concrete for sustainable construction using mineral carbonation

Approximately 8% of the man-made contributions to CO₂ emissions originate from cement production ([Winnefeld et al. 2022](#)). A promising and economical route for CM can be developed for the safe sealing of CO₂ using carbon sequestration in Ca/Mg-containing cement-based raw materials (e.g., concrete) at different stages of their lifetime or in Mg-containing silicate-based olivine rocks. During the mineral carbonation of concrete, CO₂ enters through the pores of the concrete and forms CaCO₃ while increasing the porosity by lowering the pH and replacing CH molecules. This weakens the concrete mix and reduces the overall strength

of the concrete (Balapour et al. 2018). However, the application of nano-or micro-sized silica (SiO_2) can control the carbonation kinetics by reducing the Ca(OH)_2 content, which subsequently reduces the carbonation depth in concrete to avoid damage (Lim and Mondal 2015). The simultaneous addition of nano- and micro-sized silica (10%) can reduce the carbonation depth by 33% by filling the voids between the cement grains (by micro-sized SiO_2) and pores between micro-sized SiO_2 and cement grains (by nano-sized SiO_2) (Li et al. 2017).

An economical CO_2 storage route is the production of green concrete by carbonation curing of cementitious materials obtained from ISW. Several researchers have reported that amorphous SiO_2 and nano-sized CaCO_3 produced during mineral carbonation, i.e., CMS of approximately $100\text{ }\mu\text{m}$ in size, can be a substitute for sand or additives in concrete to provide high particle strength for economical and sustainable building materials (Balapour et al. 2018; Meng et al. 2019; Wang et al. 2018). Yi et al. (2020) suggested that CMS production via a wet method to replace cement could decrease the carbon footprint of concrete. In addition, the utilization of industrial CO_2 for curing concrete can rapidly harden cement because of the reaction between CO_2 and cement clinker. A high $\text{CaCO}_3/\text{SiO}_2$ ratio ensures better CaCO_3 microcrystal nucleation and crystallization, and higher CO_2 uptake. The degree of carbonation increased by 74.2% when the optimal CMS addition ratio was 30%, and the $\text{CaCO}_3/\text{SiO}_2$ ratio was 4:1, in contrast to that of the pure cement paste. When pure CO_2 was used to cure the mixture at $40\text{ }^\circ\text{C}$ and 1.5 MPa, the strength increased by 40%, much higher than the strength obtained via natural curing (28 d) (Yi et al. 2020). Strengthening of the interfacial transition zone after CO_2 curing is thought to be the primary contributor to the increase in strength. The insights obtained from this study may help guide the sustainable construction industry in using CMS from CM.

The 2016-patented Solidia Cement is a nonhydraulic binder derived from raw materials, similar to ordinary Portland cement. However, a smaller fraction of CaCO_3 and kiln

temperature of approximately 1200 °C are used, resulting in a 30% decrease in CO₂ emissions (Chen et al. 2017). Furthermore, the Calera, Solidia, and SkyMineTM processes use hydroxides or electrolysis to increase the pH from <4 to >8 of the CO₂-equilibrated aqueous stream for carbonate precipitation, which is economically unattractive and environmentally polluting. Instead of using stoichiometric inorganic bases for alkalinity, regenerable ion exchange (IEX) materials can be a suitable alternative for shifting the pH to achieve sustainable and economical CM. H⁺ and Na⁺ ions are reversibly exchanged from a bivalent-free solution (protons are available after CO₂ dissolution at a low pH in water) and the IEX material, shifting the reaction equilibria in an alkaline CO₃²⁻ solution. This helps increase the pH and bicarbonate formation in CO₂-rich fluids to facilitate CaCO₃ precipitation (Bustillos et al. 2020). The concentrated alkaline waste solution after mineralization can regenerate the IEX materials.

4.5 Global implementation of CCS facilities in different industries

The current CCS facility portfolio appears more diverse and has been implemented in coal-based power plants, iron and steel, cement, and other chemicals and heavy industries as critical business drivers for CCS, with more comprehensive geological storage options. Carbon capture technologies are widely employed using low-cost next-generation CCS methods with safe transportation and secure injection of CO₂, without any evidence of leakage. In total, 19 CCS facilities were in operation (as of November 2019). In addition, 28 CCS units were in various stages of development, and four were under construction (Fig. 3 a). In the USA, 17 projects are under development, and 10 are in operation, primarily in the fertilizer, power, natural gas purification, hydrogen, and ethanol production industries (Kearns et al. 2021). The National Carbon Capture Center in the USA has facilities for testing new CCS-related technologies (Beck 2020).

The current CCS processing cost is high in some industries but will gradually decrease on the way to achieving carbon-neutral development by implementing newly developed CM

technologies. CCS technology is more practical than natural carbon sinks; however, natural carbon sinks are more economical ([Fig. 3 b](#)). The abscissa shows the amount of carbon sources that various carbon-emission industries, such as power, coal, prepared materials, and emissions, can replace. CCS costs are inversely proportional to the carbon concentrations in emissions, and are beneficial for CCS implementation in industry. Direct air capture (DAC) of CO₂ could have greater scope and potential in the future ([Shen et al. 2023](#)). The first substantial influx of funding for CO₂ mineralization occurred in the early 2010s. Although several CO₂-based goods have been manufactured, they have failed to garner the interest of investors who prefer to invest their money in businesses using technology with established profitable business models. Modern technological advances have led to a resurgence of interest in CO₂ mineralization, which has fueled the increased cost performance of these tools. Compared to conventional CO₂ collection methods such as absorption and adsorption, CO₂ mineralization is a potential technology with several advantages, such as mineral security and climate change mitigation ([Franks et al. 2023](#)).

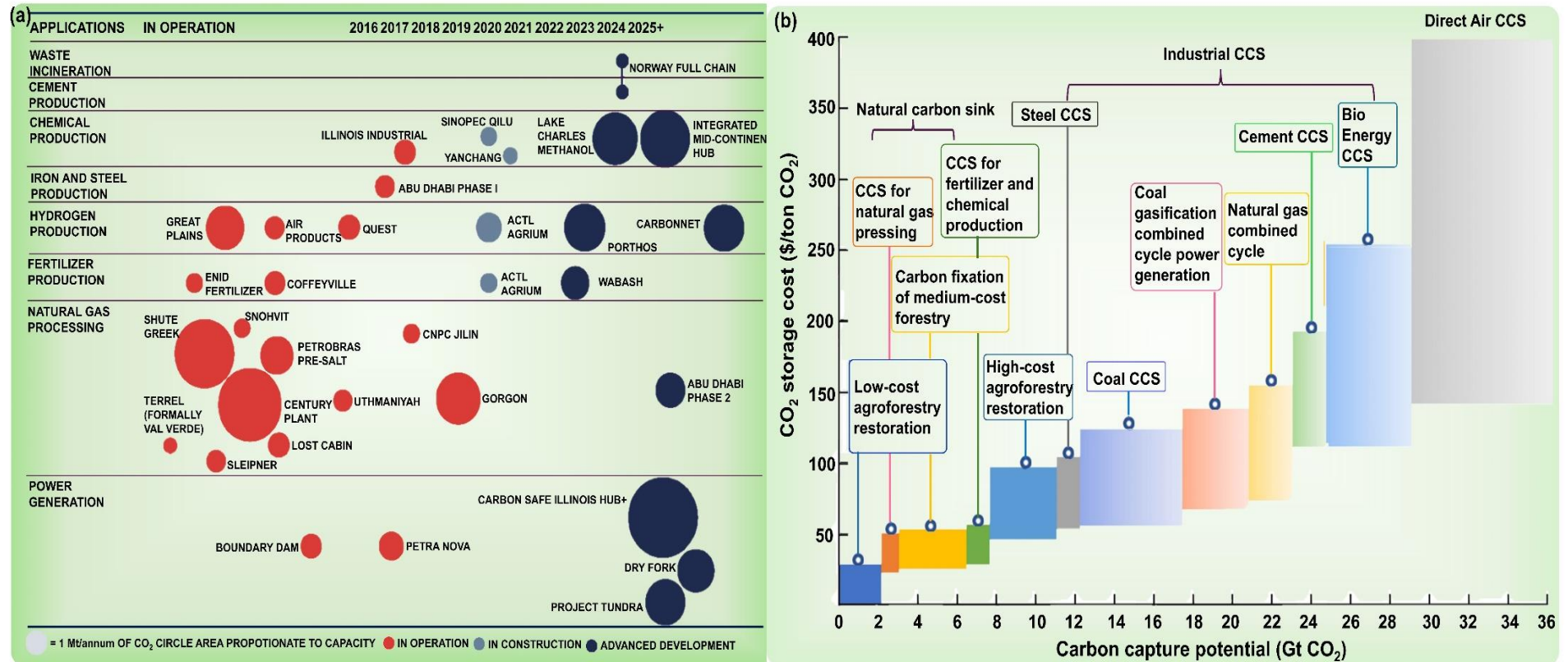


Fig. 3. Industrial scale implementation of CCS technologies, (a) Large-scale carbon capture and storage-based projects implemented by industrial applications and storage facilities, adapted from (Beck 2020) and (Kearns et al. 2021), (b) CO₂ capture potential and carbon capture and storage cost integrated with different industrial processes (source: Goldman Sachs Equity Research 2020 obtained from (Shen et al. 2023))

4.6 Current practical approaches of carbonation using industrial solid waste

Several ISWs, such as slags (from blast furnaces and steel), waste gypsum, and coal fly ash (CFA), with annual outputs of ~630 Mt, ~300 Mt, and ~1000 Mt, respectively, containing 27–50% CaO and 5–15% MgO, are useful for CO₂ sequestration. The contributions of different industrial-based alkaline solid wastes used for CO₂ mineralization (direct or indirect carbonation) worldwide are shown in [Fig. 4 a and b](#). Iron and steel slag-based mineralization can reduce the maximum direct carbonation, followed by concrete and cement waste, mining waste, and CFA. In contrast, cement and concrete waste, fly ash from coal combustion, and slag from the iron and steel industries are the top three industrial wastes with maximum CO₂ mineralization ([Pan et al. 2020](#)). The country-wise potential amounts of direct and indirect carbonation processes indicate that the top 10 countries represent ~87.1% and 89.2% of the global CO₂ reduction, respectively, and China alone contributes approximately four-fold compared to any other country ([Fig. 4 c and d](#)).

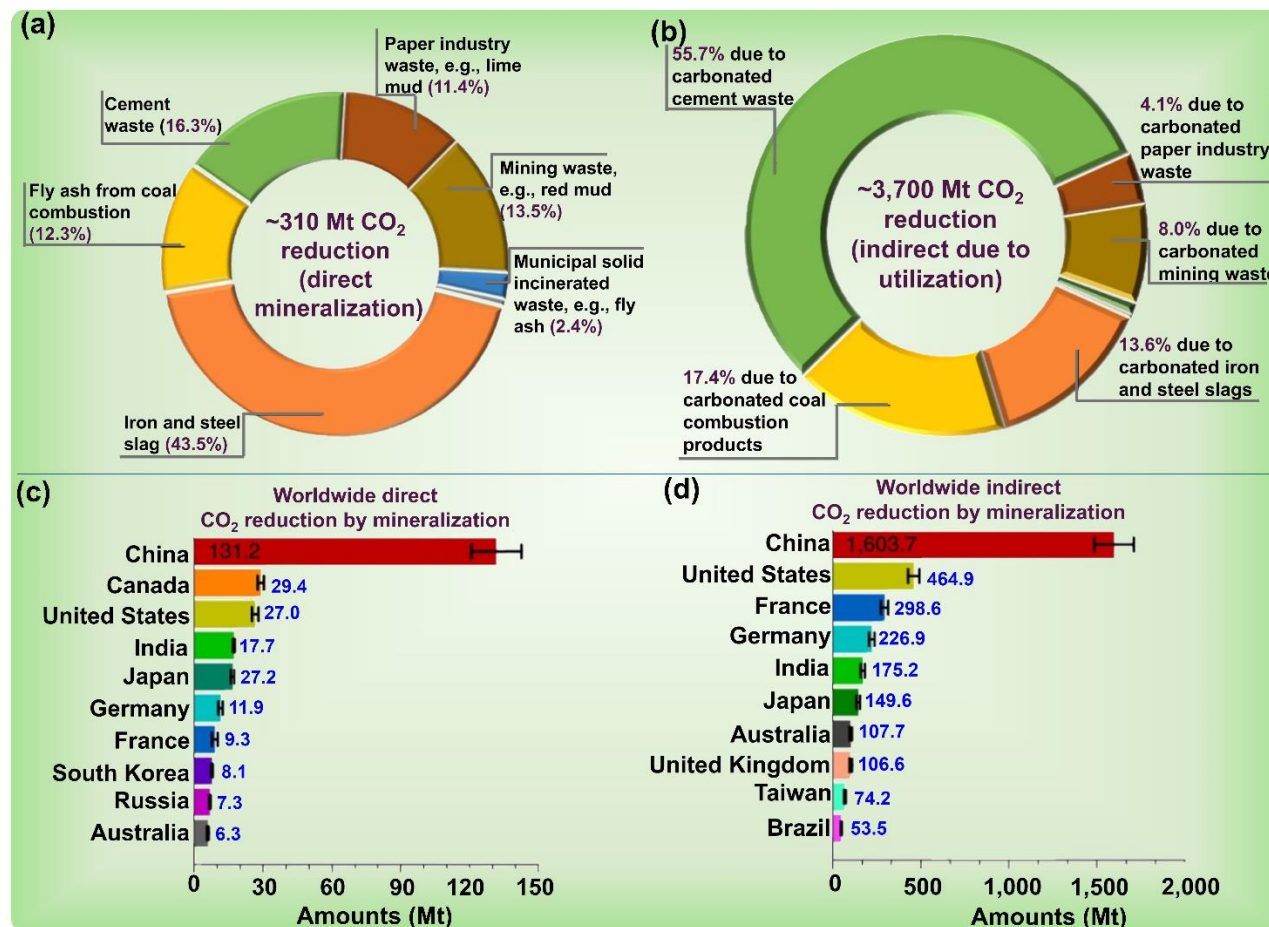


Fig. 4. CO₂ mineralization based on types of carbonation, industrial wastes, and global scenario, (a) direct mineralization using different alkaline wastes; (b) Indirect mineralization using different alkaline wastes; (c) Direct CO₂ carbonation by different countries; (d) Indirect CO₂ carbonation by different countries (adapted from (Pan et al. 2020) with permission from Springer Nature License Number 5722361231260)

4.6.1 Slag from iron and steel industries

Industrial alkaline solid wastes, such as steelmaking slag, fly ash, and bottom ash, are stabilized by rapid CO₂ sequestration using a rotating packed bed with a high-gravity facility to enhance the carbonation kinetics (Chen et al. 2020b). The solid waste (slag) from iron and steel-making industries contains 60 wt% of MgO+CaO, which indicates a large CM capacity with 0.36 tons of CO₂/ton of slag. The steelmaking industry produces 1.9×10^9 tons of steel (USGS 2020) and generates 3.8×10^9 tons of CO₂, which contributes to 7% of the global CO₂ emissions due to anthropogenic activity. Slag is a major solid waste generated from blast furnaces and steel production units. It contains Ca₂MgSi₂O₇ and Ca₂Al₂SiO₇ with considerable amounts of CaO (~40%), MgO (~10%), and Al₂O₃ (~12.5%), making it a suitable feedstock for CO₂ storage (Grubb and Berggren 2018). Dry carbonation of CO₂ using slag from a blast furnace was performed in a closed chamber for 28 days to obtain calcite with a calcium carbonation efficiency of 39% (Uliasz-Bocheńczyk and Mokrzycki 2017). Surface-modified ground granulated slag using an alkaline solution was shown to have a CO₂ carbonation efficiency 10 times higher than that of slag without alkaline pretreatment (You et al. 2011). A CM efficiency of 280 kg CO₂/ton of slag was achieved under optimized conditions of the NaCl (1 M), pressurized CO₂ flow (30 bar), and an operating temperature of 150 °C at a reaction time of 24 h (Ren et al. 2020). The salt solution aided the leaching of Ca, followed by the CM process.

Li et al. (2024) tested the pozzolanic activity of mineralized steel slag (under different CM conditions, that is, at atmospheric and high CO₂ pressures) (Li et al. 2024b). Exposure of slag to atmospheric-pressure CO₂ increased the particle size owing to the formation of calcite and expansion of Ca-silicates, sequestered 14.9% of CO₂, reduced Al and Si leaching, and increased the strength activity index by 95.8% in 28 days. The mineralized steel slag increased by 5.49% CO₂ sequestration after mixing with cement, owing to the presence of calcite minerals. In another study, the synergistic effect of carbonation curing and the addition of CO₂-

mineralized slag to cement paste was shown to enhance its comprehensive strength and CO₂ sequestration (10.76%) capability owing to microstructural and mineral composition changes (Li et al. 2024a). The low particle size slag (38 μm) enhanced the CO₂ sequestration with a high conversion efficiency of Ca at a reaction temperature of 100 °C, CO₂ pressure of 19 bar, and operating time of 30 min during direct carbonation (Huijgen et al. 2005).

4.6.2. Fly ash from coal-based power plant

Coal-based power plants can produce 750–1000 million tons of fly ash worldwide with different compositions depending on the type of coal used for combustion (Qin et al. 2019). The chemical composition of fly ash reveals the presence of MgO, CaO, SiO₂, Fe₂O₃, and Al₂O₃, with primary mineral phases of lime, magnetite, mullite, quartz, and portlandite (Meng et al. 2018). Direct carbonation processes have mainly been studied for CO₂ sequestration using fly ash as a feedstock because of its highly alkaline conditions in the presence of calcite and lime (Shao et al. 2024; Ukwattage et al. 2015). The carbonation efficiency using fly ash varies from 7.66 to 210 kg/ton, owing to variations in the CaO content of the fly ash. Direct dry carbonation is feasible because of the presence of highly reactive compounds, such as CaO and Ca(OH)₂, which can be operated under modest (25 to 200 °C) (Dananjayan et al. 2016; Mazzella et al. 2016) or high (>600 °C) temperatures (Liu et al. 2018b) with slow reaction kinetics. Direct aqueous carbonation, especially in salt brine (ammonium chloride), can accelerate the reaction rate using fine particles of fly ash (Hosseini et al. 2016). Ji et al. (2018) proposed a hybrid technology for simultaneous CO₂ absorption using amine-based solvents followed by mineralization using fly ash where CO₂ migrated from the solvent to precipitate as CaCO₃ at a mild temperature of 40 °C with default solvent regeneration.

4.6.3 Waste gypsum

Industrial solid gypsum wastes (red gypsum, phosphogypsum (PG), and gypsum generated during flue gas desulfurization) primarily contain CaSO₄·2H₂O with 32.6 wt% of CaO that can

mineralize 0.26 kg CO₂/ton of waste. High carbonate reactivity with CO₂ at atmospheric pressure and ambient temperature was shown by CaSO₄·2H₂O, with 100% conversion of Ca into CaCO₃ (Lee et al. 2012; Song et al. 2014). The waste gypsum obtained from the desulfurization of flue gas is comparatively pure and is primarily used to produce economically valuable CaCO₃ precipitates. Ammonia dosing was found to be a more critical factor than the CO₂-flow rate and solid/liquid ratio in driving dissolved CaCO₃ toward impurity-free CaCO₃ precipitates (Song et al. 2014). During direct aqueous carbonation, high concentrations of alkaline solutions are added to make the process economically unattractive because of the high cost of the applied alkalis (i.e., NH₄OH or NaOH) compared to that of crystallized salts (e.g., Na₂SO₄ or (NH₄)₂SO₄) (Azdarpour et al. 2018; Pérez-Moreno et al. 2015). To avoid foreign impurities, an inorganic acid (H₂SO₄) is used as a lixiviant to extract Ca from waste gypsum during indirect carbonation (Rahmani 2020). Direct aqueous multi-phase systems expedite the reaction rate owing to the direct contact of reactants in gas-liquid-solid systems. However, trace amounts of radionuclides in PG and red gypsum are a major concern for their application as feedstock for the CM process.

4.6.4. Mining waste

Another feedstock that could be an alternative to the geological storage of CO₂ through mineralization is mining waste, which contains considerable amounts of Mg, Ca, and Fe(II) silicates. Industries consume enormous quantities of metals resulting from mining operations, which is inevitable for the economic development of any country. This results in the generation of a huge volume of mine waste (2 to 6.5 billion tons/year) annually (Renforth et al. 2011). Furthermore, metal mining generates 0.419 billion tons of ultramafic and mafic waste (Power et al. 2013). The metal and diamond mining industries directly contribute to approximately 3.6 Gt of CO₂ emissions (Azadi et al. 2020). Mining waste (metal and diamond commodities) is a potential feedstock for the CM process to offset CO₂ emissions from mining industries (Paulo

et al. 2021). The potential for CO₂ storage in these mines is substantial and can counteract 1.5% of global CO₂ emissions annually (Punia 2021; Shih et al. 1999). Minerals with different dissolution rates, such as wollastonite, forsterite, brucite, lizardite, serpentine, and diopside, were identified as cation sources. Brucite had a higher dissolution rate over a wide pH range than Mg-bearing silicate minerals. However, other Mg silicates, such as serpentine, can quickly release loosely bound Mg for effective CO₂ mineralization (Stubbs et al. 2022).

Bullock et al. (2021) estimated that approximately 1.1 to 4.5 Gt CO₂ might be captured annually using waste generated by mafic and ultramafic rock-hosted operations and a high amount of Cu-hosted deposits, which is 31% to 125% of the mining industry's primary emissions. Hariharan and Mazzotti (2017) investigated the ability to predict the CO₂ mineralization efficiency of partly dehydroxylated lizardite particles at low temperatures (≤ 90 °C) and low CO₂ partial pressures (≤ 1 bar). This process is advantageous for efficient CO₂ emissions management and hazardous material remediation, such as the recovery of chrysotile from mining waste for CM (McCutcheon et al. 2014). However, the low dissolution rate of minerals is a major obstacle in natural CO₂ capture; therefore, only a fraction of the potential, that is, only 3%–21%, may be achieved in < 50 years. The mining waste is ground to small particle sizes of 50 to 200 μm or even 4 μm to enhance carbonation efficiency (Garcia et al. 2010). The expedition of natural carbonation can be increased by increasing the temperature (150 to 180 °C) and pressure (150 bar) during *ex-situ* aqueous carbonation. The heat pretreatment of serpentine-type rock converts it into olivine at 550 to 650 °C by removing adsorbed water molecules followed by partial or complete dihydroxylation to enhance the carbonation reaction rate. The passivating layer built around the reacting particles is a major barrier to the aqueous carbonation of silicate particles. Enhancement of the porous layer and limiting the formation of the passivation layer over silicates can be achieved through a proper understanding of the carbonation mechanism.

4.7 Technology providers for large-scale carbon mineralization

Scaling up and practical demonstrations of CO₂ mineralization from ISW, such as gypsum, mine waste, steel slag, blast furnace slag, and CFA, are key steps forward. Wang and Maroto-Valer (2011) reported a pH-swing method using an aqueous solution of (NH₄)₂SO₄ to extract Mg-rich solution from serpentine rock for carbonation with the regeneration of ammonium salts (Fig. 5 a). A two-stepped CO₂ mineralization process is developed at Åbo Akademi University that uses ammonium salts for extraction of Mg as Mg(OH)₂ from serpentine minerals in the first step followed by its carbonation in the dry phase in a second phase (Romão et al. 2014) (Fig. 5 b). Only a handful of China's CO₂ mineralization and usage projects are in the proof-of-concept stage. For example, with the help of Sichuan University, the Sinopec Group accomplished a pilot project to directly mineralize CO₂ using PG from exhaust gas (100 Nm³/h) (Falzone et al. 2021). Hence, it is crucial to consolidate practical knowledge and theoretical rules supporting laboratory-based research on unit operation acceleration, continuously improve the process steps, scale up and incorporate the process, and encourage technological development and implementation. The Sinochem Chongqing Fulong Chemical Co., Ltd. demonstration facility produces 1×10⁵ tons of concentrated PG annually. The project improved CO₂ mineralization by transforming the Ca-sulfate in PG into CaCO₃ and (NH₄)₂SO₄ using an ammonia medium and the absorption of high-concentration CO₂ from the ammonia synthesis method (Fig. 5 c). The final products were a fertilizer called (NH₄)₂SO₃ and light CaCO₃ of high purity (>99%) (Quinn and Sahu 2019). The phosphate rock was carbonated at a rate of more than 97%, and more than 95% of the CO₂ produced was used. Several researchers have attempted to recover soluble K resources and mineralized CO₂ simultaneously from naturally occurring potash ores, such as K-feldspar (KAlSi₃O₈) and ISW (PG) (Gan et al. 2016; Wang et al. 2014; Xie et al. 2015). The coupling process was investigated under optimal conditions, such as activation of K-feldspar and PG (mass ratio of 1:2) at 1200 °C for 2 h,

followed by CO₂ mineralization at 40 bar partial pressure and 100 °C temperature to obtain maximum K extraction (as K₂SO₄) and CO₂ mineralization (CaCO₃) ratios of 87% and 7.7%, respectively (Fig. 5 d) (Wang et al. 2014).

Another pilot-scale demonstration project by the Sinopec Group, in collaboration with Sichuan University, used PG as feedstock to mineralize CO₂ from flue gas (Xie et al. 2015) (Fig. 5 e). An ammonia-rich solution was used to absorb CO₂ from the flue gas to convert ammonium carbonate to reduce the CO₂ concentration from 15% to 4.5% in the flue gas, which was subsequently scrubbed with an acidic PG slurry in acid pickling to form (NH₄)₂CO₃. The ammonia-free CO₂ slurry was further reacted with fresh PG in a three-phase reactor to form CaCO₃ with CO₂ capture and conversion efficiencies of 75% and 92%, respectively. At the California pilot plant, mineral carbonation was tried using seawater and alkaline waste. Solid aggregates were produced after dewatering cementitious material, the carbonation product of brine solution and alkaline industrial waste, following the Calera process (Fig. 5 f). However, the requirement of a considerable amount of energy for processing seawater and industrial waste for carbonation limited the wide application of the Calera process (Zaelke et al. 2011)

Furthermore, a CCS demonstration project was constructed by Yuanchu Technology Corporation in conjunction with Tsinghua University using chemical looping and mineralization using silicate ore and building solid waste (Meng et al. 2021). In this study, Ca-containing silicates and construction debris were chlorinated to produce calcium chloride, which was combined with ammonia, water, and CO₂ to undergo a carbonation process that converts CO₂ into CaCO₃. The system is intended to handle 1,000 tons per annum of CO₂ from flue gas released from a cement factory with 90% mineralization efficiency and a 50% net absorption rate. An industrial experiment on CO₂ deep mineral carbonation curing was conducted in collaboration with Zhejiang University and Henan Qiangnai New Materials Co., Ltd. Many existing steam curing systems have been modified, and a pilot-scale project has

demonstrated a novel CO₂ remedial process (Simonetti et al. 2019). This initiative consumed 10,000 tons of CO₂ annually during CO₂ curing of 1×10⁸ MU15 solid concrete and lightweight bricks without availing the energy-intensive and polluting procedures used during conventional steam curing. Recently, many laboratory-scale studies have been conducted on CM using ISW at the laboratory scale (Iizuka et al. 2012, Lin et al. 2024, Liu et al. 2021). Several studies have been conducted at the pilot scale in continuous mode to target industrial-level operations (Iizuka et al. 2017, Kemache et al. 2017, Pan et al. 2013, Said et al. 2016). A number of obstacles are encountered during the scale-up of technologies, such as limitations in heat or mass transfer during the enlargement of operating vessels, resulting in lower efficiency of the process (Ren et al. 2021). A pilot-scale plant with a capacity of 200 L was developed at Aalto University in 2014 for indirect carbonation using slag waste feedstock (from steel industries) and NH₄Cl as the alkaline agent (Said et al. 2016). A pilot-scale reactor was fabricated to operate a batch of 190 L of alkaline solution (1 M NH₄Cl) and 20 kg of slag waste to extract Ca (~80%) within 1 h. The temperature was controlled at <45 °C during indirect carbonation to prevent evaluation of the NH₄Cl solution, which was recycled back after 71% precipitation of CaCO₃ to make the process economical.

Laboratory- and pilot-scale studies have been conducted on carbonation using bauxite solid waste under atmospheric (long-term experiments) and pressurized (short-term experiments) CO₂ flows (Han et al. 2017). The reaction rate was accelerated by adding CaCl₂ to achieve 83 kg of CO₂ storage per ton of bauxite following a process similar to natural carbonation, which facilitates smooth field implementation. Several lab- and pilot-scale projects were implemented based on direct and indirect mineral carbonation for industrial applications, using fly ash from coal plants, iron and steel slag, waste gypsum, and other Ca/Mg-containing residues (Table 2). Moderately concentrated CO₂ extracted from anthropogenic sources reacts with several industrial wastes or feedstocks containing Ca/Mg to

be converted into Ca/MgCO_3 for commercial applications, such as aggregates and additional cementitious materials, for global carbon mitigation.

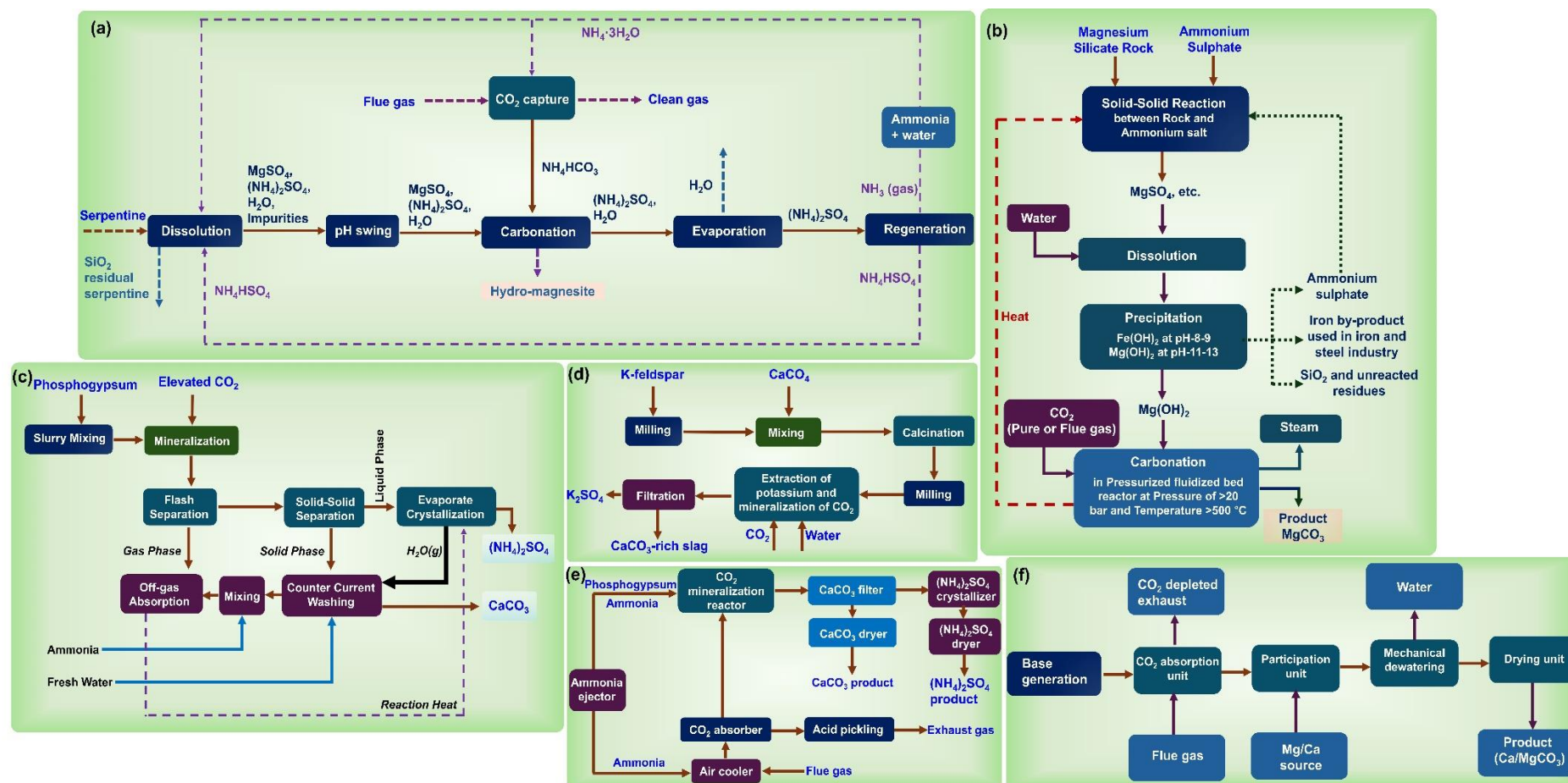


Fig. 5. Schematic process-flow of CO₂ mineralization via (a) pH swing method with recycling of ammonium salt (adapted from (Wang and Maroto-Valer 2011)); (b) Åbo Akademi University route (Romão et al. 2014); (c) Phosphogypsum-based high-concentration CO₂ (from ammonia synthesis process) mineralization procedure developed by Sinochem and the Chinese Academy of Sciences (Li et al. 2018); (d) Thermally activated K-feldspar and phosphogypsum starting feedstocks (Wang et al. 2014); (e) Phosphogypsum-based low CO₂ (from flue gas) concentrations mineralization developed by Sinopec and Sichuan University (Xie et al. 2015); (f) Calera process (Zaelke et al. 2011)

Table 2. Pilot-and lab-scale studies for carbonation using industrial solid waste as feedstocks

Carbonation process	Feedstocks	System details	Operating conditions	Yield or recovery	Remarks (pros and cons)	References
Direct carbonation	Fly ash from incineration of wastepaper (source of Ca, Pb, and Ba)	- Pilot-scale natural carbonation set-up	- Maintained 0.3 L/kg of liquid to solid ration - Left seven days for natural carbonation	- Carbonates conversion of Ca, Pb, and Ba - Achieved final pH of 11.4 after 90 days of natural carbonation	Pros - Pilot-scale technology was developed for safe disposal of fly ash by natural carbonation of heavy metals (Ba and Pb) - Immobilized Pb and Ba into carbonates to decrease their concentrations by 98% and 98.5%, respectively - Percolated rainwater after carbonation can be disposed of without treatment Cons - The carbonation process is relatively slow - It requires a humidification and frequent homogenization step	(Bouzar et al. 2024)
	Mining waste (Ca source), NaCl (0.7 M),	- Conducted Carmex large-scale project implemented in France (2009-2012) using ultramafic mining waste - Autoclaves of 2 and 3 L capacity with magnetic stirrer	- Carbonation tested in an autoclave reactor at CO ₂ pressure of 10 to 20 bar, 0.7 M NaCl solution, the reaction time of 24 to 96 h at 180 °C	- Achieved 80% of carbonation efficiency	Pros - Utilization of mining waste for mineral carbonation without heat activation using flue gas - Coupling of mechanical exfoliation and carbonation in an autoclave reactor enhanced CO ₂ sequestration Cons - The process needs high pressure - Mechanical exploitation is an energy-intensive process	(Bodénan et al. 2014)
	Bauxite residue and waste gypsum obtained from the desulfurization of flue gas (Ca source), CaCl ₂ (lixiviant), atmospheric CO ₂	- Laboratory and field-scale testing - Used bench-scale of 800 mL glass reactor - Tested field-scale set-up spread in a 4 m ² area using 100 kg of bauxite	- Tested for 55 days in bench-scale bauxite residue neutralization during carbonation using different concentrations of CO ₂ bubbling, 100%, 14%, and 1% - Pressurized CO ₂ flow of 1 L/min for short-term neutralization in 100 g/L of bauxite suspension	- Maximum 83 kg of CO ₂ sequestered per ton of bauxite	Pros - Addition of waste gypsum or CaCl ₂ improved CO ₂ sequestration - Due to improper mixing of bauxite residue during field study, the resulting slow reaction rate - The treatment reduces the chemical dosing that is required for neutralization of bauxite Cons - Less CO ₂ sequestration efficiency without ca-source	(Han et al. 2017)

Serpentine Tailings (Mg source) and flue gas from the cement industry (CO ₂ source)	- Used 18.7 L capacity Parr reactor	- Long-term neutralization using atmospheric CO₂ in 50 and 16 g/L solid suspension of 1:1 ratio of bauxite and gypsum waste - Field testing of 100 kg bauxite neutralization at atmospheric pressure of CO₂ with and without CaCl₂ solution tested - Used solid/liquid ratio of 150 g/L and stirring speed of 600 rpm at 22 to 25 °C - Flue gas containing 12-20% of CO₂ was circulated at 2 to 10 atm pressure at a residence time of 15 to 30 min	- Captured 90-96% of CO₂ from flue gas that captured 80 to 250 kg of CO₂ per ton of solid residue at laboratory scale	- Neutralization with atmospheric CO₂ requires a large land area and a relatively slow process Pros - Recovered high purity MgCO₃·3H₂O using flue gas of cement plant - Leached 19% of Mg - Relatively less captured of total dissolved CO₂ (only 25-33%) at pilot-scale - Heat treatment for Mg leaching and its precipitation as carbonate by dissolved CO₂ were two crucial factors in achieving maximum yield Cons - Mineral carbonation needs to be dissolved CO₂ under 2 to 4 bar pressure at high stirring condition - Achieved a relatively low carbonation rate (25-33% of dissolved CO₂) using mining waste and flue gas	(Kemache et al. 2017)
Industrial alkaline solid residue, coal fly ash (CFA)	- Lab-scale Stainless steel (SS) reactor of 1 L capacity with four baffles, pressure-gauge, port and inlet valve - Maximum working pressure of 10 bar at ambient temperature	- CFA and 100% pure CO₂ in SS reactor at 10 bar pressure for 1 h (for dry route) - Slurry of CFA water ratio of 15, at 4 bar pressure for 2 h, stirred at 900 rpm in SS reactor (for wet route)	- Captured 26.3 g CO₂/kg of CFA (dry route) - Captured 50.3 g CO₂/kg of CFA (wet route)	Pros - Shown a simple and direct carbonation of CFA-containing CaO (~10%) - Achieved higher carbonation efficiency by wet carbonation (67% conversion efficiency) compared to dry one Cons - Carbonation efficiency influenced by process parameters and direct carbonation routes	(Dananjay an et al. 2016)

Blast-furnace slag	- A lab-scale jacketed autoclave of 200 mL capacity with stirring capacity at high pressure - Maximum operating pressure of 60 bar	- Addition of NaCl (1 M), pressurized CO₂ flow (30 bar), and 150 °C of operating temperature at a reaction time of 24 h during CO₂ mineralization	- 280 kg CO₂/ton of slag waste at 10 (S/L ration), 30 bar of CO₂ pressure, and 150 °C of temperature	Pros - CO₂ deposition over blast-furnace slag for CO₂ reduction and utilization of slag - Expedite CO₂ mineralization by adding 1 M NaCl solution in slurry of slag - Achieved higher dissolution of Ca²⁺ from slag in saline condition Cons - Required high temperature and pressure led to high operating cost	(Ren et al. 2020)
CFA from coal-based power plant	- Used a lab-scale reactor of 500 mL capacity made up of SS along with a reference reactor using only CO₂ - Maintained temperature by keeping the reactor in a thermostatic bath	- Homogenized CFA with a particle size of 0.212 mm at 7.5 bar pressure and at 45 °C of operating pressure	- Achieved carbonation of 182 g CO₂/kg (dry route of direct carbonation)	Pros - Achieved maximum carbonation rate of 74% depending upon initial CaO content in CFA - Required mild operating conditions for direct dry carbonation to store CO₂ in CFA Cons - Homogenized particle size, high pressure, and moderate temperature increase the operating cost	(Mazzella et al. 2016)
Red gypsum	- Used a lab-scale SS autoclave of 100 mL capacity with maximum operating temperature and pressure of 450 °C and 200 bar, respectively - Reactor facilitated with stirring	- Prepared red gypsum slurry by using 10 g of gypsum with particle size <45 µm in 50 mL deionized water - Stirred at 1000 rpm - Experiment conducted at operating pressure and temperature of 20 bar at 25 °C, respectively, at an operating time of two hours - Used red gypsum of	- Achieved 2.85% and 8.94 % maximum Fe and Ca carbonation efficiency	Pros - Contained Ca and Fe in red gypsum for carbonation - Produced carbonates of Ca and Fe during direct wet carbonation Cons - Needs further research to improve the carbonation efficiency - High operating pressure makes the process costly	(Azdarpour et al. 2018)
Electric arc furnace (EAF) and basic oxygen	- Carbonation experiment conducted in pilot scale jacketed Permex autoclave	- Slags pretreatment, shaping products between 75 and 609 bar, and CO₂ curing in an autoclave at 50 to 100	- Achieved carbonation of 180 – 200 g CO₂/kg of slag	Pros - Used pretreated and carbonated slags for CO₂ capture and storage during accelerated mineral carbonation	(Quaghebeur et al. 2015)

	furnace (BOF) slags	of 104 L capacity with automatic temperature and CO ₂ pressure control systems	bar and 20 – 140 °C for 16 h in CO ₂ enriched environment		- Production of high-quality construction material with blast furnace and SS slags - Produced <i>in-situ</i> carbonated materials that acted as binder for cementing the slag particles Cons - Required high operating costs due to operating under high temperature and pressure	
	Solid waste from the concrete and cement sector (Ca source), flue gas (CO ₂ source), and NH ₄ NO ₃ (aq) (weak acid used for leaching)	- A pilot plant of 200 L that can handle 160 kg/h of solid	- Weakly acidic solution of NH₄NO₃ was used to leach Ca from solid waste from the concrete sector in a pilot-scale study - Clear solution containing Ca was used for CO₂ absorption to form pure CaCO₃	- The leachate was used to store 0.9 kg of CO ₂ /h to form CaCO ₃	Pros - Used concrete waste as a Ca source for sequestering CO₂ from flue gas - High recyclability NH₄CO₃ was used for leaching of Ca from waste concrete for CO₂ mineralization into cementitious material - Residual lixiviant NH₄NO₃ is recycled and reused, making the overall process economical Cons - Low dissolution of Ca during leaching of waste concrete - Need solvent recovery step to make the process economical - Spillage and evaporation of ammonia from lixiviant corrode the equipment and contaminate the environment	(Meijssen et al. 2023)
Indirect aqueous carbonation	Steel slag (Ca source), NH ₄ Cl (lixiviant)	- Fabricated pilot-scale reactor of 200 L capacity at Aalto University, Finland	- A batch of 20 kg solid waste (slag) and 190 L of an alkaline solution of 1 M NH₄Cl was used to leach 80% of Ca in 1 h in a pilot-scale study - Temperature was maintained below 45 °C - Residual NH₄Cl was recycled	- Carbonation efficiency of 71%	Pros - Pilot scale operation to utilize steel slag as feedstock to sequester CO₂ into CaCO₃(s) - Regeneration and recycling of solvent (NH₄Cl) makes the process cost-effective Cons - Ammonia-based corrosive solvent was used for leaching of Ca from steel slag - High temperature (60 °C) caused evaporation of ammonia gas	(Said et al. 2016)
	Concrete sludge (Ca source), boiler gas for CO ₂ from pole production	- Tested in a pilot-scale plant in Japan in a 40 m ³ reactor	- Extracted Ca from concrete sludge using water in 1:1 ratio at ambient temperature in 6 h in a pilot-scale study	- Sequestered 140 kg of CO ₂ using 356.7 tons of sludge	Pros - Achieved net CO₂ reduction of 118 kg from 140 kg of CO₂ and 3.6 ×10⁵ ton of concrete sludge - Operated without purification and pressurized boiler gas as a CO₂ source during carbonation	(Iizuka et al. 2017)

plant, and water (lixiviant)		- Carbonated by blowing boiler gas in ca-solution for 3 h and left for one week		- Achieved industrial-grade high-purity (>97%) CaCO_3 Cons -	
Slag from the blast furnace (Ca and Mg source), $(\text{NH}_4)_2\text{SO}_4$ (ammonia source), and water as the leaching agent	- Used rotary kiln (self-designed) for continuous indirect carbonation using slag	- Roasted at 350 to 450 °C the mixture of $(\text{NH}_4)_2\text{SO}_4$ and slag - Absorbed tail gas in water to form ammonia water, which was used for CO_2 capturing as $(\text{NH}_4)_2\text{CO}_3$ or NH_4HCO_3 - Roasted product was leached in water to obtain the sulfates of Mg and Ca	- Achieved a net reduction of 36 kg CO_2 after concealing energy consumption during the process	Pros - Lab to pilot scale caused several problems, such as limited heat and mass transfer and kiln ring formation - The data of the study may guide for pilot or commercial scale application - Need proper optimization of process parameters Cons - Released wastewater after mineral carbonation - Relatively slow carbonation process due to slow leaching of Ca in water as solvent	(Ren et al. 2021)
Solid waste lizardite, after heat activation in a kiln (Mg source) and NaHCO_3	- Pilot scale batch reactor of 30 L capacity was used	- Raw lizardite was milled in a ball mill and sieved to obtain 75 μm particle size, followed by drying in an oven at 110 °C for 24 h to produce magnesite by mineral carbonation in a pilot plant of 30 L plant capacity - Heat treated in a kiln at 630 °C for 4 h - A slurry was prepared in 300 mL distilled H_2O by adding 56 g lizardite (heat treated) and 0.64 M of NaHCO_3 - Slurry of 15 wt.% was treated in a reactor vessel at 250 °C and 200 bar	- Magnesite yield of 35% was achieved using 150 bar CO_2 pressure, 15% solid concentration in the presence of NaHCO_3 (0.64 M)	Pros - Yielded 35% higher magnesite in the pilot batch study compared to laboratory scale due to improved mixing - Observed no significant changes by using either distilled water or tap water - Adding NaHCO_3 during carbonation favors magnesite yield Cons - Required chemical reagent (NaHCO_3) to achieve higher yield - High operating cost due to an energy-intensive process, such as heat activation of feedstock (lizardite) and high operating pressure (200 bar) and temperature (250 °C) during mineral carbonation in SS vessel	(Benhelal et al. 2018)
Slag from the basic oxygen furnace and	- Used rotating packed bed reactor in Taiwan to	- Fed 30% containing CO_2 flue gas along with slag and wastewater in	- Achieved 98.3% of CO_2 sequestration	Pros - Specifically suitable for iron and steel industries to capture and store CO_2 utilizing its by-products,	(Pan et al. 2013)

wastewater from the cold rolling unit (Ca source), flue gas from the hot stove (CO ₂ source)	capture CO ₂ from flue gas using alkaline solid and liquid wastes from steel industries	20 mL/g at a rotating speed of 1000 rpm for 1 min and operating temperature of 25 °C in a rotating packed bed reactor	i.e., wastewater, slag, and flue gas (containing 30% v/v of CO ₂) in rotating packed bed reactor - Technology has the potential to be adopted by steel plants across the world to extract 96-99% of CO ₂ from flue gas to convert CaCO ₃ at ambient temperature and pressure in short operating time Cons - Dissolution of CO ₂ was the rate-limiting step for mineral carbonation - Required concentrated CO ₂ -containing flue gas for its capture and storage
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Although long-known, carbon capture technology has not been widely adopted because of its high implementation costs and other barriers. Several leading companies in the field of carbon capture have advanced and perfected their methods to provide safe, efficient, eco-friendly, and scalable solutions. Several companies and startups have started CCS processes using patented technologies to make efforts to rescue the planet. The patent-based technologies that were successfully implemented at start-up and industrial scales are shown in [Table 3](#). Start-up companies related to CCS play a critical role in developing novel and commercially viable CCUS technologies in compliance with the increasing pressure to achieve net-zero emissions targets. The developed technologies use a low-cost CO₂ removal process, either by direct carbon capture using train braking energy (CO₂Rail) or by developing a two-step CCS process that starts with the separation of CO₂ flue gas or biogas in a dilute stream, followed by purification through a membrane or amine solution via absorption (Rushnu Inc.). The captured anthropogenic CO₂ may be delivered to end users to be used as feedstock for a circular economy, or compressed and stored in an underground reservoir for natural mineralization. These CCS-based industries are pleading for global CO₂ reduction efforts from existing large-point sources of emissions, which would help solve issues related to the existing historical carbon in the atmosphere. One option is to invest in carbon capture firms, even though the industry remains nascent ([Subramanian and Madejski 2023](#)). Market instability is caused by high technology prices, supply chains, and competition intensity. The CO₂ mineralization techniques and their applications are summarized in [Table 4](#).

Table 3. CO₂-capturing using breakthrough technologies practiced in companies and startups.

Companies or startups	Carbon capturing technology	Foundation year	CO ₂ capture potential	Key points	Sources
CO ₂ Rail Company, Oberlin, Ohio, USA	- Direct Air Capture (DAC) rail-based technology facilitated with microprocessor-controlled CO₂ sorbent media - Potentiation of molecular captors helps in the desorption of CO₂ - Uses heat energy during regenerative braking system to capture CO₂ from air - Specially designed rail lines outfitted with battery array, CO₂ capture systems, compression equipment, ancillary gear that utilize energy during braking, and solar panels (without external charging) fixed on rail cars - Funnelling diesel locomotive exhaust as a point CO₂ source to mitigate carbon emission. - The collected CO₂, liquified, stored, and can be offloaded and delivered as feedstocks for end users or injected into underground sequestration	2021	- The technology may achieve one Gt scale of CO₂ removal potential using train brake energy by utilizing available rail infrastructure	- Economic process for decarbonization using railway operation - It has a modular and scalable approach - A minor adjustment in existing locomotive infrastructure needs to be implemented in the technology - The CO₂ collection does not block the landscape - Rail DAC of CO₂ based on adsorbent or absorbent material that must be highly efficient in sorption and desorption - Recycling of sorbent material should be simple and economic - Extraction of CO₂ from air is an energy-intensive process due to its dilute concentration	(Bachman et al. 2022)
Rushnu Inc., carbon transformation Solutions (Pleasanton, United States)	- CO₂ capture from the dilute gas stream from heavy industries and utilization into valuable products through one-step technology - Utilizes solvent-based absorption process for selective separation and storage of CO₂ followed by mineralization into minerals applied in building materials, glass industry, adhesives, and paints industries - Technology used for biogas purification to remove CO₂ and enrich methane content	2022	- Developed a compact modular unit with carbon capturing capacity of thousands of tons CO₂	- Produces valuable minerals by CO₂ mineralization used in different industries (glass, building materials, paints, and adhesive industries) - Accelerated and stable carbon mineralization using mantle peridotites - It is in the initial stage that needs field scale implementation	(Hanifzadeh 2021)

Blusink Ltd. (London, United Kingdom)	- It uses a patented material that provides a settling zone for algae (coralline) larvae inside Seawater - Algae remove CO₂ by calcifying it into pink rocks, i.e., rhodolith beds for the habitat of reef fish - The reef fish contribute to the restoration of the reef and the habitat of living beings inside Seawater.	2022	- Facilitates biological carbon capture with four-times carbon uptake efficiency compared to terrestrial carbon mineralization	to assess the efficiency of energy consumption, the extent of carbon capture and storage, and techno-economic feasibility aspects - Technology helps to restore and rewild ecosystems in oceans - Low-cost and affordable technology utilizing widely available materials - Long-term geological sequestration without any risk of re-emission of captured carbon - Injecting CO₂ into the ocean may affect the marine ecosystem due to a change in pH	(Neira 2022)
Zeru.Earth (London, United Kingdom)	- Developed new technologies for CO₂ removal utilizing biochar, Seaweeds, weathered rocks	2022	- Implemented several projects based upon direct air capture (Octavia Carbon), Seaweed farming (Seaweed generation), and increased rock weathering (UNDO)	- Deploying direct air capture machine in Kenya as Octavia Carbon project to provide a cost-effective solution to carbon removal from air by utilizing >85% of process energy through Kenya's natural heat - A car size Octavia Carbon machine can absorb CO₂ equivalent to a forest spread over 25 acres at the cost of 1,000 USD/ton of CO₂ - The operating cost relatively higher for CO₂ capture and storage	(Contreras and Arráiz 2022)
Carbon Blade Corporation	- Developed distributed direct air capture technology using carbon-blade for absorbing CO₂ from the air by circulating electrochemical solvents	2021	- Capacity to sequester 1 Mt of CO₂ per day	- It has zero waste and utilizes 100% renewable energy - Cost-effective, 100 USD/ton of CO₂ capture	(Roberts et al. 2021)

(San Diego, USA)	- It is a loop electrochemical system with self-generated air and energy supplied				- No need for transportation costs for CO ₂ - The technology based on electrochemical solvent-based CO ₂ adsorption that requires desorption and regeneration of solvent	
Blue Skies Minerals GmbH (Berlin, Germany)	- Aqueous solution of CO ₂ produced by direct air capture process mixed with mine tailing (ground mine waste) with additives to convert carbonated minerals	2023	- Estimated carbonation target of 1 to 3 Mt of CO ₂ per year		- The process reduces metal leaching for carbonation from mining wastes to make an economical carbonation process - Dissolution of CO ₂ in aqueous phase determines its carbonation rate in mine tailings	(Sobolewski and Silko 2022)
CCU International (Aberdeen, United Kingdom)	- The technology used to capture CO ₂ directly from mixed gases or flue gases - Flue gases pass through multi-stepped processes under a pressurized silica gel - After several depressurization steps, gases are released except CO ₂ , which is later collected for storage or utilization	2022	- The company provides different capacities of carbon capturing machine that varies from 0.3 tons of CO ₂ /day to 100 tons of CO ₂ /day		- Supply system that can capture CO ₂ from small and large industries, such as oil and gas, petrochemical, manufacturing, rail, shipping, breweries, and anaerobic digester - It has a compact design with a small footprint, is easy to scale up, and is cost-effective - The process requires the multi-stepped process to extract CO ₂ from flue gas using silica that needs desorption and regeneration	(Sharma and Doherty 2022)
Carbon8 Systems Ltd. CO ₂ ntainer™ (U.K.)	- Developed CO ₂ ntainer™-based on accelerated carbonation technology for CO ₂ capture and conversion into artificial limestone - CO ₂ reacts with solid waste effluent from industries, such as steel slag and cement waste - Prepared aqueous slurry by adding water to alkaline waste and mixed with liquid CO ₂ during carbonation	2006	- Captured 0.1 to 0.2 tons of CO ₂ per ton of aggregates		- Innovative, compact, and modular CO ₂ ntainer™ technology can seamlessly integrate with industries for CCS process - Used industrial alkaline waste (steel slag and thermal waste	(Carey and Hills 2006)

CarbonCure (Canada)	- Prepared round-shaped aggregates using binder and fillers (e.g., lime, silica, and alumina) in pelletizers at 90 °C - Method developed to increase strength and accelerate concrete curing by CO₂ injection to reduce 25% (lower down 95 to 118 min) and 23% (lower down 103 to 126 min) of time for initial and final setting - Reaction between moisture cement and CO₂ causes a decrease in pH, shrinkage, and induction corrosion due to carbonation - Improved absorption, strength, freeze-thaw function, and permeation of chloride due to early-stage carbonation of precast concrete	2012	- Carbonation capacity of 1.6×10⁵ Mt CO₂	- from the cement industry) for CCS to produce carbonated materials - Produce new materials CarbonCure ready mix using CCS technology using cement and concrete waste that can be replace cement cost by 3-5% without reducing concrete strength - Reduced 7% to 8% of cement addition by CarbonCure technology without compromising the strength of the concrete mix, thus reducing 150 kg of CO₂ per m² of ready-mix concrete utilization	(Niven 2012)
Solidia Technologies (USA)	- Low lime-containing nonhydraulic cement (mainly composed of rankinite and wollastonite minerals) results in 30% less carbon emissions - A relatively lower calcination temperature of 1200 °C is used for Solidia cement compared to 1600 °C for Portland cement, which further reduces 30% carbon emission due to less fuel consumption - Lower carbon emission, i.e., 550 kg CO₂ per ton of Solidia cement compared to 800 kg CO₂ per ton of Portland cement - Patented process is used for concrete production using fine and coarse aggregates, CO₂, and Solidia cement formed calcite polymorphs and silica gels during carbonation	2008	- Capacity 250-300 kg of CO₂/ton of cement - Sequestering capacity of 1.5 GT of CO₂/year	- Technology developed to produce less carbon footprint concrete by reducing temperature for Solidia cement manufacturing that led to decrease a 30% less CO₂ emission from kiln - Solidia cement directly interacts with CO₂, whereas conventional Portland cement interact with CO₂ in presence of water - Curing of Solidia cement sequestered 290 to 310 kg of CO₂/ton of cement	(Hill 2007)
Blue Planet Systems (San	- Developed Ca-riched fine and coarse aggregates following natural mineral carbonation	2012	- Captured 0.44 tons of CO₂/ton aggregates	- Used different types of solid wastes (steel slags, demolished concrete waste, bauxite, fly ash,	(Constantz 2012)

Francisco, USA)	- Produced concrete aggregates mineralized CO₂ to form a structure which becomes carbon-negative - Reduces 0.1 ton of CO₂ emissions by using one ton of concrete - Bay Aggregates at San Francisco fabricates its first plant to manufacture carbon-neutral concrete using alkaline solid waste from Medanos Energy Center following Blue Planet patented technology			- kiln dust from cement industries) for CCS process - A pilot-scale plant launched by Blue Planet in collaboration with Sulzer Chemtech Ltd. to make sustainable cement industry by reducing CO₂ emissions	
Calera Corporation (USA)	- Forms stable CaCO₃ by mixing CO₂ from flue gas to Ca or Mg riched solution - At the demonstration plant set up by Calera in California (Moss Landing), can mineralized CO₂ 30,000 tons/year, equivalent to carbon emissions from a 10-MW power plant - Approximately 5.5×10⁵ tons of cementitious materials can be produced using flue gas produced from a 100-MW power plant using Calera technology	2007	- Plant capacity of 3.4 Mt of CO₂ per year and 0.46 tons of CO₂ per ton of alkaline waste materials - TRL, 8-9	- Supplied CaCO₃ precipitates to the cement industry as supplementary materials (cementitious materials) - Carbon mineralization offsets the CO₂ emissions from cement industries and earned revenue	(Ginder-Vogel 2007)
CarbiCrete (Canada)	- It is a Montreal-based CO₂ reduction company that develops carbon-neutral construction technology - Mineralized steel slags using CO₂ to produce cementitious materials to reduce or replace cement utilization for cement-free concrete - Developed a specialized chamber for concrete curing using CO₂ to obtain full strength concrete after 24 h	2016	- Captured 0.2 tons of CO₂ per day	- Carbon-negative concrete production starts at Lafarge Canada by using CarbiCrete technology by making a partnership in 2023 to handle sustainability issues in the cement industries	(Mahoutian and Stern 2016)

Table 4 Techniques, sources, and applications of CO₂ mineralization

CO ₂ mineralization technique/sources	Applications of CO ₂ mineralization	Reference
Coupled reaction-extraction-crystallization technique	- Simulated distiller waste conversion for the industrialization process	(Hariharan and Mazzotti 2017)
Sea water desalination brine	- Reduction of CO ₂ to carbonate mineral under the coexistence of Ca and Mg	(Chen et al. 2017)
Unavoidable CO ₂ source to CO ₂ sink	- The cement industry is based on CO ₂ mineralization	(Wang et al. 2021)
Utilization of steel slag	- The establishment of a waste-to-resource supply chain	(Ostovari et al. 2021, Pan et al. 2017)
Brine electrolysis process	- In the production of sodium carbonate, hydrogen, and chlorine	(Lee et al. 2018)
Wet process	- CO ₂ mineralization slag is a sustainable construction material in the industry	(Wang et al. 2021)
Coupling CO ₂ mineralization and solvent extraction	- To investigate the precipitation of magnesium from a high Mg/Li ratio brine	(McCutcheon et al. 2014)
High-gravity carbonation process	- Refining slag with calcium-containing wastewater for CO ₂ mineralization	(Chen et al. 2020a)
Saline water-based mineralization pathway	- Gigaton-Scale CO ₂ management: reducing atmospheric CO ₂ levels	(La Plante et al. 2021)
Ex-situ mineral carbonation	- The reaction of Ca-, Fe-, and Mg-silicate minerals with gaseous CO ₂ to form geologically stable, naturally occurring solid carbonate minerals	(Gerdemann et al. 2007)
Neutralization process of acid mine drainage for CO ₂ mineralization	- Ca/Mg containing acid mine drainage can be used to sequester CO ₂ by forming Ca/MgCO ₃ through CO ₂ injection	(Lee et al. 2016)

Application of CO ₂ for leaching and precipitation to extract metals by CO ₂ mineralization	- CO ₂ plays a dual role in the selective leaching of lithium from spent LiFePO ₄ cathode materials, followed by its precipitation into Li ₂ CO ₃	(Xu et al. 2023a)
CO ₂ mineralization using brine source and fly ash	- CaO-riched fly ash mixed with brine solution to increase the pH to increase the carbonation efficiency during CO ₂ mineralization of Ca from both sources to CaCO ₃	(Soong et al. 2006)
Electrolytic carbonation of Ca present in fly ash	- Synergistic approach for the remediation of various wastes, such as fly ash from coal-fired plants, brine solution, and CO ₂ . Electrolysis of brine solutions increases acidity to liberate Ca from fly ash, which can be carbonated using CO ₂ to produce high-purity CaCO ₃	(Lu et al. 2016)
Aqueous carbonation using fly ash obtained from the incineration of municipal solid waste and cold-rolling wastewater	- Fly ash and rolling wastewater could sequester 102 g per kg of solid ash with a conversion efficiency of 90.7% to CaCO ₃ while consuming 180 kWh of energy for fixing one ton of CO ₂	(Chang et al. 2015)

5. Carbon capture and storage: Evaluation of feasibilities and implications

The development and refinement of multiple CCS technologies are crucial for reducing CO₂ and alleviating global climate change. The selection and application of these highly impactful tools require evaluation of their practical applications and consideration of anticipated challenges that strongly depend on various factors, including types of industry (for direct carbonation), infrastructure (e.g., grid and transportation), geology and geography (for indirect carbonation or geological sequestration), and the political stands of local governments and/or nations. Although accurate assessments must consider tangible and intangible factors, this section focuses on the most common factors that play key roles in determining techno-economic feasibility and environmental and societal challenges ([Roy et al. 2023](#)).

The environmental effects of the CCS and CM methods must be determined through a meticulous evaluation of the final CO₂ emissions using LCA ([Zimmermann et al. 2020](#)). It is crucial to maintain direct or indirect carbon emissions lower than fixed CO₂ levels during mineral carbonation ([Wang et al. 2024b](#)). Additionally, the global warming potential (GWP) was used to compare the global warming impacts of various gases, which measures the amount of energy that would be absorbed during the emission of one ton of gas relative to the equivalent amount of CO₂ over a given period ([US EPA 2024](#)). The CCS associated with power plants can decrease the GWP by 63–82%, with maximum reductions obtained for oxy-fuel combustion in finely powdered coal integrated with gasification plants, and minimum reductions by post-combustion capture integrated with gas turbine plants. In contrast, only a 4–48% reduction in GWP can be achieved for mineral carbonation during CO₂ utilization ([Cuéllar-Franca and Azapagic 2015](#)).

[Nduagu et al. \(2012\)](#) reported environmental implications, such as material requirements, energy consumption, and emissions of CO₂ equivalents (CO₂-eq) during the mineralization of one ton of CO₂ from a coal power plant and silicate rock (Mg-source) using

LCA. The multi-stage mineralization route developed in Finland (Åbo Akademi University) was used to extract Mg from the ore to form $\text{Mg}(\text{OH})_2$, followed by MgCO_3 formation in a fluidized bed reactor under high temperature and pressure. The process required 3.6 GJ of energy and generated 517 kg CO_2 -eq for one ton of CO_2 mineralization into MgCO_2 owing to the high process heat requirement. A consumption of 300 kWh per ton- CO_2 of energy was required for the sequestration of 160 kg CO_2 per ton of slag by extracting Ca in NH_4Cl solution (pH swing mineralization) from the slag from the iron and steel industry, followed by precipitation to CaCO_3 at 80 °C with 45% conversion (Kodama et al. 2008). The economic feasibility of the CM technology is a major challenge in promoting the utilization of ISW for CM (Kirchofer et al. 2012). During the production of value-added products from electric arc furnace slag using indirect aqueous carbonation, reagents (NaOH , NaCl , CH_3COOH , and HNO_3) cost 600–4500 USD per ton of CO_2 sequestration (Sanna et al. 2014).

Thonemann et al. (2022) reported a negative median global warming impact for direct aqueous carbonation, indirect solid carbonation, and carbonation curing through a detailed meta-LCA, resulting in various pessimistic, realistic, and optimistic scenarios (Fig. 6 a). Categorization into optimistic, realistic, and pessimistic scenarios depended on the end-product utilization (Ca-/Mg-carbonates and silica). In optimistic scenarios, end-products should replace conventional products, whereas in realistic scenarios, MgCO_3 is considered a valuable end product. In pessimistic scenarios, end-products are not used. A high negative value of -35 kg CO_2 -eq per kg of CO_2 used for the global warming impact was calculated for carbonation mixing in a scenario to avoid concrete production. The positive impact of indirect aqueous carbonation on global warming can be attributed to its high thermal energy consumption (Thonemann et al. 2022).

CM technologies, particularly direct aqueous and indirect solid carbonation, use diluted CO_2 (without purification or flue gases) or pure concentrated CO_2 (after purification). One

potential benefit of using dilute CO₂ is that no purification steps are required. However, a reduced carbonation yield is expected with dilute CO₂. Furthermore, a lower global warming impact was observed for indirect solid carbonation using concentrated CO₂ instead of dilute CO₂ in the pessimistic, realistic, and optimistic scenarios. In contrast, direct aqueous carbonation yields similar results in the optimistic scenario (Fig. 6 b). The low conversion rate and lower global warming impact of CO₂ purification technologies result in an overall higher global warming impact when using dilute CO₂.

Geological CO₂ storage by injecting CO₂ is a resilient, feasible, safe, and secure option, as suggested by the estimation and quantitative prediction of longer stays followed by mineralization toward permanent trapping (Alcalde et al. 2018). The risk assessment of CO₂ release due to faults, release from wellbores, leakage from pressurized pipelines, and transportation can be mitigated by monitoring the injection process and deploying sensing technologies at the surface and subsurface. Monitoring geological carbon storage helps detect leakages, reduces the amount of CO₂ loss, and quantifies leak sizes. Furthermore, simulations can help improve our understanding of the fundamentals and models of different aspects of the storage process. This enriches the knowledge on pressure build-up in pipelines, fluid flow, and geo-level chemical and mechanical changes to avoid CO₂ loss at storage sites (Ajayi et al. 2019; Rodosta and Ackiewicz 2014). The Safe Drinking Water Act regulates the geological storage of CO₂ by injection via the Environmental Protection Agency (EPA) to avoid any changes in subsurface water chemistry due to contamination. An EPA advisory was issued to regulate this by categorizing it into six classes of wells for geological storage to protect against potable water contamination. Class VI wells are used during geological storage by injecting CO₂, and are identified as Areas of Review to be monitored for footprints to protect drinking water (Nicot et al. 2009). It has been suggested that CO₂ injection must be performed below the maximum

penetration of wells to compensate for the uncertainty of the long-period geological storage of CO₂ and migration of the CO₂ plume in the well ([Yu et al. 2023](#)).

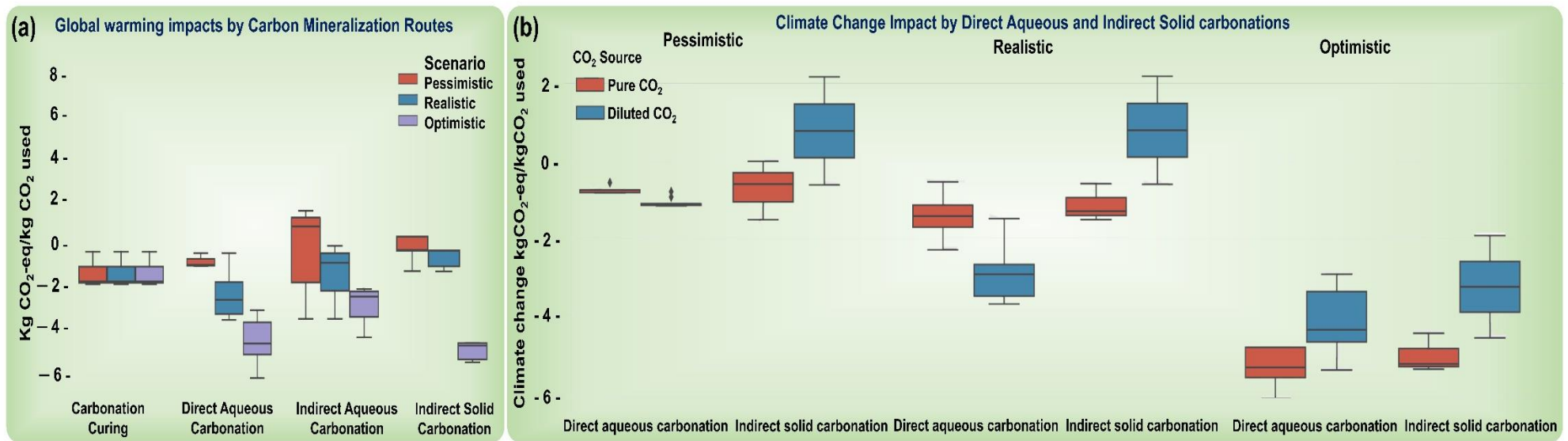


Fig. 6. Life cycle assessment of different methods of CO₂ mineralization, (a) Global warming impact by CO₂ mineralization techniques, (b) Effect on climate change due to direct aqueous and indirect solid carbonation methods depending on the CO₂ sources and scenarios (adapted from (Thonemann et al. 2022))

The implementation of LCA has resulted in a shift in how environmental impact assessments are conducted (Ghasemi et al. 2017). Previously, environmental impact assessments were restricted to mundane processes, such as manufacturing and processing; however, they have now been expanded by including the entire life cycle of the product supply chain. Companies can better control the environmental effects of their entire product supply chain using LCA-based environmental impact assessments. The energy penalty produced by the process intensification of a system may strike a balance between the energy and environmental implications of the CM processes (Thonemann et al. 2022). Local energy distribution should be appraised for the environmental implications of power and heat generation. Determining net CO₂ emissions is crucial because CO₂ mineralization by ISW is a technical option for reducing the effects of climate change. Raw material production may result from CO₂ emissions, offset the ISW treatment requirement, and produce CO₂ credits from trace mineralization at the margins. Thus, by using ISW-based CO₂ mineralization technology, it is possible to account for the CO₂ released. For example, construction blocks made from mineral carbonation of steel slag have been subjected to LCA (Jia et al. 2019). Carbonated blocks have compressive strengths comparable to those of the traditional Portland-cement-based building blocks. However, carbonated blocks exhibit fewer adverse environmental effects. Lee et al. (2020) proposed an indirect carbonation method involving mineral carbonation of steel slag to produce nano-sized CaCO₃. The project was economically viable, and the LCA revealed a net decrease of 8% in CO₂ emissions compared with that of the traditional approach. The carbon footprints were calculated using LCA by Ostovari et al. (2020) for seven indirect and direct carbonation methods. Products derived by carbonating natural minerals and steel slag are used as partial cement substitutes, and all approaches reduced climate effects between 0.44 and 1.17 tons of CO₂-eq per ton of CO₂ stored.

6. Outlook of scientific challenges and perspectives in CO₂ mineralization

Despite considerable advancements in the mineralization of anthropogenic CO₂ into Ca and Mg carbonates, several scientific challenges need to be addressed. Naturally occurring minerals or ISWs used as alkaline materials have heterogeneous chemical compositions. The iron and silica constituents of alkaline sources affect the rate of carbonate formation and its mechanisms in different fluidic environments. Thus, predicting the kinetics of carbonate formation using inconsistent multicomponent feedstock is challenging. In a multiphase environment, the carbonation process converts anthropogenic CO₂ into carbonates. Future research should focus on industrial waste to achieve suitable operating conditions and CM routes for specific wastes to exploit their full carbon-storage potential.

Different characterization methods, such as X-ray and neutron scattering, tomography, and spectroscopy at operating temperatures and pressures, are necessary to understand the details of the fluid chemistry, structure, and morphology during carbonate formation and its underlying mechanisms. Advanced characterization of carbonated materials is required to find suitable applications, such as algal feed or construction materials, depending on their composition, stability, and strength.

Among the different CM routes, direct aqueous carbonation and indirect carbonation are primarily used at the laboratory scale. Nevertheless, indirect carbonation yields high-purity Ca/MgCO₃, which involves the recycling of leaching solvents, and should be considered. The development of novel solvents or adsorption materials is required to capture and supply CO₂ and recycle it *in-situ* after carbonate formation, as a suitable substitute for the acid/base consumption process to produce Mg- or Ca-carbonates. Novel synthetic pathways are required to produce nano-or mesoscale Ca-or-Mg carbonates by utilizing anthropogenic CO₂ in a few additional steps using regenerable sorbents or solvents ([Werner et al. 2014](#)).

The integration of CM with different pathways for selective resource recovery from industrial and other waste sources, such as brine solutions from desalination and oil recovery

plants, cement waste, lime mud, AMD, asbestos residue, fly ash, and waste LiFePO_4 batteries, can demonstrate commercially viable and effective carbon reduction strategies. However, a fundamental understanding of the mechanisms of dissolution and carbonate formation rate in silicate minerals, as well as the formation of proper crystal shapes and high-purity carbonated products requires future research on *ex-situ* CO_2 storage in the feedstocks mentioned above.

The extraction of REEs from natural leachate (AMD), CFA, and high-TDS brine solutions obtained from oil field extraction or desalination plants using flue gas to produce carbonates of Ca, Mg, and Na through the CM process can be a synergistic approach to address various wastes. However, the commercialization of direct brine mining requires proper evaluation of its composition, source, and generation path after treatment and optimization of the operating process parameters, such as pH, CO_2 flow rate, and temperature, for high conversion efficiency through carbonation reactions ([Kumari et al. 2024](#)).

CMS from ISW can be used as a substitute for cement during carbonation curing to produce green concrete as a sustainable building material by reducing the carbon footprint. The application of nano- and micro-sized SiO_2 along with CO_2 -sequestered CaCO_3 positively affects the durability of concrete with rapid CO_2 curing by reducing the carbonation depth and improving the intensification by filling the pores among the cement grains ([Yi et al. 2020](#)).

Recently, an integrated approach using CO_2 as a leaching agent for the selective recovery of lithium, followed by precipitation into Li_2CO_3 from spent LiFePO_4 batteries, has shown great potential as a CM for decarbonization, along with energy and resource recovery ([Xu et al. 2023a](#)). However, the relatively slow dissolution and precipitation of lithium is a key hindrance that can be overcome by further research on gas-liquid crystallization methods to obtain high-quality Li_2CO_3 for sustainable lithium recovery and recycling of waste resources ([Kim et al. 2024](#)).

To promote carbon reduction efforts, the advancement of scientific knowledge in the CM process in natural subsurface environments and artificially engineered systems is essential. Moreover, the balanced action of the 3Es, that is, engineering (smaller number of unit operations), economic (economically feasible operating processes), and environmental (eco-friendly approaches), should be evaluated when developing novel processes for utilizing industrial alkaline waste for the CM process in future research.

7. Conclusions

In this review, we examine the current state of CO₂ mineralization technologies and discuss the possibilities for future expansion. Several alternative strategies have been proposed in different countries to reduce CO₂ emissions. In addition, methods successfully adopted in the CCS industry were comprehensively reviewed. The storage of CO₂ in water-insoluble Ca/MgCO₃ is thermodynamically favorable for CM. Efforts to expedite CM at low temperatures and CO₂ concentrations in industrial flue gas streams present considerable practical challenges during mineral dissolution and carbonate precipitation, and the reaction rate is unfavorable. CO₂ separation and mineralization via the looping of solvents (especially amine-based) between carbon-loaded and free states can accelerate the capture and storage of CO₂ in Ca- and Mg-carbonates as a single-step, low-temperature operation, and adaptable pathway. CM has shown the potential to replace traditional CCS methods for reducing carbon through *in-situ* or *ex-situ* leakage-free mineral sequestration. Several critical aspects of CM are discussed with respect to its historical background, potential future developments, carbonation mechanisms, technological options, practical uses, integrated carbonation approaches that utilize different wastes, and environmental impacts. Efforts to reduce CO₂ emissions through CM have provided opportunities to improve several types of manufacturing processes. The pressing need to reduce and utilize CO₂ through CCS may contribute considerably to climate change mitigation. The selection and application of CCS technologies require a robust

evaluation of their techno-economic feasibility and environmental impact over extended time and length scales. The development of numerous cutting-edge and environmentally friendly CCUS technologies, as well as their integration into manufacturing industries and the power sector may result in a clean, green, and sustainable planet in the future.

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